STAT 332 Sampling and Experimental Design

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Contents

1 Sampling

Lecture 1, 2025/09/03

1.1 Intro & Survey Terminology

Assignments are there to help you prepare for exams!

Why study survey sampling?

- Obtain data for a question of interest from a population.
- Cannot take a census.
- Finite time and resources.

Definition 1.1 (**Target Population**).
The complete collection of individuals we wish to study and draw conclusions about. Denoted by $U = \{1, 2, \ldots, N\}$.

Definition 1.2 (**Sampled (Study) Population**).
The collection of individuals that could be included by our sampling design.

Definition 1.3 (**Sampling Unit**).
The object being sampled.

Definition 1.4 (**Frame**).
The list of all sampling units.

Definition 1.5 (**Observation Unit**).
The object on which we take measurements.
Note. Often, this is the person we ask questions to.

Three main steps at which error and bias can be introduced in the conducting of a survey:

- Constructing the frame (coverage error or study error)

- Obtaining the sample/respondents (sampling bias/error, non-response error)
- Measurement of the response (measurement error)

Definition 1.6 (**Sampling Design (Sampling Protocol)**).
A **sampling design** (or sampling protocol) is the mechanism by which we select our samples.

Note. **Sampling error** is the estimation error due to sampling variability.

Definition 1.7 (**Selection Bias**).
Differences between the target population and the effective sampled population.

Note. Ideally, target population = population that can be effectively sampled.

Sources of selection bias:

- Coverage: e.g., can be difficult to construct a complete frame
- Sampling: e.g., improper sampling design
- Non-response: people don't respond or are not reachable

Definition 1.8 (**Measurement Error**).
The measured values of the attributes on the observation units are different from the true values.

Lecture 2, 2025/09/08

1.2 Probability-Based Sampling Design

Assume that frame is complete (ideal case). Assume also that $U = \{1, 2, \dots, N\}$. We denote the responses (corresponding values for attribute of interest) by $y_1, y_2, \dots, y_N$.

Population quantities:

- $\mu = \frac{1}{N} \sum_{i \in U} y_i$ (population mean)
- $\tau = \sum_{i \in U} y_i$ (population total)
- $\sigma^2 = \frac{1}{N-1} \sum_{i \in U} (y_i - \mu)^2 = \frac{1}{N-1} \left[\sum_{i \in U} y_i^2 - N\mu^2\right]$ (population variance b/c population is finite).

Note.

$$\begin{aligned}
\sigma^2 &= \frac{1}{N-1}\sum_{i\in U}(y_i-\mu)^2 \\
&= \frac{1}{N-1}\sum_{i\in U}(y_i^2-2\mu y_i+\mu^2) \\
&= \frac{1}{N-1}\left[\sum_{i\in U} y_i^2 - 2\mu\sum_{i\in U} y_i + N\mu^2\right] \\
&= \frac{1}{N-1}\left[\sum_{i\in U} y_i^2 - 2N\mu^2 + N\mu^2\right] \\
&= \frac{1}{N-1}\left[\sum_{i\in U} y_i^2 - N\mu^2\right].
\end{aligned}$$

Suppose we have a sample $s \subseteq U$, a subset of frame with sample size n.

Estimates of population quantities based on sample:

- $\hat{\mu} = \frac{1}{n}\sum_{i\in s} y_i$, sample mean estimates population mean
- $\hat{\tau} = \sum_{i\in s} y_i$, sample total estimates population total
- $\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i\in s}(y_i-\hat{\mu})^2$, sample variance estimates population variance

Note. As in STAT 231, we treat estimates as constant values based on a specific sample, i.e. y_i's are constants. We treat the sampling design/process as random, so s is a realization of a random variable S.

Sampling design tells us $\mathbb{P}(S=s)$ which captures the **inclusion probabilities**.

Definition 1.9 (Inclusion Probabilities).

$$\pi_i = \mathbb{P}(i \in S), \quad \pi_{ij} = \mathbb{P}(i, j \in S), \quad i \neq j \in U.$$

Then, we can define estimators as functions of S:

- $\tilde{\mu} = \frac{1}{n}\sum_{i\in S} y_i$ (sample mean estimator)
- $\tilde{\tau} = \sum_{i\in S} y_i$ (sample total estimator)
- $\tilde{\sigma}^2 = \frac{1}{n-1}\sum_{i\in S}(y_i-\tilde{\mu})^2$ (sample variance estimator)

Definition 1.10 (**Simple Random Sampling (SRS)**).

SRS is also known as **SRS without replacement (SRSWOR)**. Take a sample of size n out of N, such that all possible samples of size n are equally likely.

Note. The randomness in SRS comes from the sampling design, not the responses.

Example. Let $N = 4$, $n = 2$, then there are $\binom{4}{2} = 6$ possible samples:

$$\{1,2\} \quad \{1,3\} \quad \{1,4\} \quad \{2,3\} \quad \{2,4\} \quad \{3,4\},$$

each with probability $\frac{1}{6}$, e.g. $\mathbb{P}(S = \{1,2\}) = \frac{1}{6}$.

Question: How many possible samples of size n?

Answer: $\binom{N}{n} = \frac{N!}{n!(N-n)!}$.

Thus, we have

$$\mathbb{P}(S = s) = \frac{1}{\binom{N}{n}} \quad \text{for any subset } s \text{ of size } n.$$

We can also compute inclusion probabilities:

$$\pi_i = \mathbb{P}(i \in S) = \frac{\text{\# of samples with unit } i \text{ in it}}{\text{total \# of possibilities}} = \frac{\binom{1}{1}\binom{N-1}{n-1}}{\binom{N}{n}} = \frac{n}{N}.$$

Note. $\binom{1}{1}$ is the number of ways to choose unit i (to include i, we must pick i, so 1 way), and $\binom{N-1}{n-1}$ is the number of ways to choose the remaining $n-1$ units from the other $N-1$ units.

$$\pi_{ij} = \mathbb{P}(i, j \in S) = \frac{\text{\# of samples with units } i, j \text{ in it}}{\text{total \# of possibilities}} = \frac{\binom{2}{2}\binom{N-2}{n-2}}{\binom{N}{n}} = \frac{n(n-1)}{N(N-1)}.$$

Note. $\pi_{ij} \neq \pi_i \pi_j$ because sampling from a finite population without replacement means that the events $i \in S$ and $j \in S$ are not independent.

Lecture 3, 2025/09/10

R Intro & SRS in R

Here is some basic syntax for R:

```
x <- c()    # creates an empty vector
for (i in 1:50) {
x[i] <- 2*i     # fill it with some stuff
}
x   # look at what it now has
mean(x)
var(x)  # sample mean and variance
```

Obtaining SRS in R:

```
> N <- 10
> n <- 5
> x = c(1:10)
> sample(x,5, replace=F)    # replace=F indicates without replacement
[1] 6 2 4 7 8
> sample(x,5, replace=F)
[1] 8 4 5 6 3
```

Probability Review

Let X be a discrete random variable. Then,

(1) $\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x), \quad \mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x).$

(2) $\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2.$

(3) Linearity: $\mathbb{E}\left[\sum_i^n a_i X_i\right] = \sum_i^n a_i \mathbb{E}[X_i].$

(4) $\mathrm{Var}\left(\sum_i^n a_i X_i\right) = \sum_i^n a_i^2 \mathrm{Var}(X_i) + 2\sum_i^n \sum_{j \neq i} a_i a_j \mathrm{Cov}(X_i, X_j).$

(5) $\mathrm{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\,\mathbb{E}[X_j].$

(6) If X_i and X_j are independent, then $\mathrm{Cov}(X_i, X_j) = 0$.

Example. Let's define an indicator random variable. Let

$$I_i = \begin{cases} 1, & \text{if unit } i \text{ is selected in the sample,} \\ 0, & \text{otherwise.} \end{cases}$$

Then, $\mathbb{P}(I_i = 1) = \mathbb{P}(i \in S) = \pi_i$ and $\mathbb{P}(I_i = 0) = 1 - \pi_i$. Thus,

$$\begin{aligned} \mathbb{E}\left[I_i\right] &= 0 \cdot (1 - \pi_i) + 1 \cdot \pi_i = \pi_i, \\ \mathbb{E}\left[I_i^2\right] &= 0^2 \cdot (1 - \pi_i) + 1^2 \cdot \pi_i = \pi_i, \\ \operatorname{Var}(I_i) &= \mathbb{E}\left[I_i^2\right] - (\mathbb{E}\left[I_i\right])^2 = \pi_i - \pi_i^2 = \pi_i(1 - \pi_i). \end{aligned}$$

For $i \neq j$, consider

$$I_i I_j = \begin{cases} 1, & \text{if both } i \text{ and } j \text{ are selected in the sample,} \\ 0, & \text{otherwise.} \end{cases}$$

Then, $\mathbb{P}(I_i I_j = 1) = \mathbb{P}(i, j \in S) = \pi_{ij}$ and $\mathbb{P}(I_i I_j = 0) = 1 - \pi_{ij}$. Thus,

$$\begin{aligned} \mathbb{E}\left[I_i I_j\right] &= 0 \cdot (1 - \pi_{ij}) + 1 \cdot \pi_{ij} = \pi_{ij}, \\ \operatorname{Cov}\left(I_i, I_j\right) &= \mathbb{E}\left[I_i I_j\right] - \mathbb{E}\left[I_i\right] \mathbb{E}\left[I_j\right] = \pi_{ij} - \pi_i \pi_j. \end{aligned}$$

So, we can write $\tilde{\mu}$ in terms of indicators:

$$\tilde{\mu} = \frac{1}{n} \sum_{i \in S} y_i = \frac{1}{n} \sum_{i \in U} I_i y_i = \frac{1}{n} \sum_{i=1}^{N} I_i y_i.$$

Definition 1.11 (Bias, Unbiased Estimator).

Suppose that $\tilde{\theta}$ is an estimator for θ. Then, the **bias** of $\tilde{\theta}$ is

$$\operatorname{Bias}\left(\tilde{\theta}\right) = \mathbb{E}\left[\tilde{\theta}\right] - \theta.$$

$\tilde{\theta}$ is an **unbiased estimator** for θ if $\operatorname{Bias}\left(\tilde{\theta}\right) = 0$, i.e. $\mathbb{E}\left[\tilde{\theta}\right] = \theta$.

Question: What are $\mathbb{E}[\tilde{\mu}]$ and $\mathrm{Var}(\tilde{\mu})$ for SRS?

Answer:

$$\begin{aligned}\mathbb{E}[\tilde{\mu}] &= \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{N} I_i y_i\right] = \frac{1}{n}\sum_{i=1}^{N} y_i \underbrace{\mathbb{E}[I_i]}_{\pi_i} \\ &= \frac{1}{n}\sum_{i=1}^{N} y_i \frac{n}{N} = \frac{1}{N}\sum_{i=1}^{N} y_i = \mu.\end{aligned}$$

So $\tilde{\mu}$ is an unbiased estimator for μ.

For SRS,

$$\mathrm{Cov}\left(I_i, I_j\right) = \pi_{ij} - \pi_i \pi_j = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}.$$

Then,

$$\begin{aligned}\mathrm{Var}(\tilde{\mu}) &= \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{N} I_i y_i\right) \\ &= \frac{1}{n^2}\left[\sum_{i=1}^{N} y_i^2 \underbrace{\mathrm{Var}(I_i)}_{\pi_i(1-\pi_i)} + 2\sum_{i=1}^{N}\sum_{j\neq i} y_i y_j \,\mathrm{Cov}\left(I_i, I_j\right)\right] \\ &= \frac{1}{n^2}\left(1-\frac{n}{N}\right)\frac{n}{N}\left[\sum_{i=1}^{N} y_i^2 - \frac{1}{N-1}\sum_{i=1}^{N}\sum_{j\neq i} y_i y_j\right]\end{aligned}$$

Note that $\sum_{i=1}^{N}\sum_{j\neq i} y_i y_j = \sum_{i=1}^{N} y_i \sum_{j\neq i} y_j = \sum_{i=1}^{N} y_i(N\mu - y_i) = N\mu\sum_{i=1}^{N} y_i - \sum_{i=1}^{N} y_i^2 = N^2\mu^2 - \sum_{i=1}^{N} y_i^2.$

$$\begin{aligned} &= \frac{1}{n^2}\left(1-\frac{n}{N}\right)\frac{n}{N}\left[\sum_{i=1}^{N} y_i^2 - \frac{1}{N-1}\left(N^2\mu^2 - \sum_{i=1}^{N} y_i^2\right)\right] \\ &= \frac{1}{n^2}\left(1-\frac{n}{N}\right)\frac{n}{N}\frac{1}{N-1}\left[((N-1)+1)\sum_{i=1}^{N} y_i^2 - N^2\mu^2\right] \\ &= \frac{1}{n^2}\left(1-\frac{n}{N}\right)\frac{n}{N}N\frac{1}{N-1}\left(\sum_{i=1}^{N} y_i^2 - N\mu^2\right) \\ &= \left(1-\frac{n}{N}\right)\frac{\sigma^2}{n}.\end{aligned}$$

Thus for SRS, $\mathbb{E}[\tilde{\mu}] = \mu$ and $\mathrm{Var}(\tilde{\mu}) = \left(1-\frac{n}{N}\right)\frac{\sigma^2}{n}$, where $\left(1-\frac{n}{N}\right)$ is called the **finite population correction (fpc)**.

To get CIs, we estimate $\mathrm{Var}(\tilde{\mu})$ with $\widehat{\mathrm{Var}}(\tilde{\mu}) = \left(1-\frac{n}{N}\right)\frac{\hat{\sigma}^2}{n}$, as it turns out that $\mathbb{E}\left[\hat{\sigma}^2\right] = \sigma^2$.

Lecture 4, 2025/09/15

Exercise: Show that $\mathbb{E}\left[\hat{\sigma}^2\right] = \sigma^2$ (PS2).

In practice, we need to estimate $\operatorname{Var}(\tilde{\mu})$ because we don't know σ^2.

Question: What is an unbiased estimator of $\operatorname{Var}(\tilde{\mu})$?

Answer: Note that $\mathbb{E}\left[\left(1-\frac{n}{N}\right)\frac{\tilde{\sigma}^2}{n}\right] = \left(1-\frac{n}{N}\right)\frac{\sigma^2}{n}$ (hint for A1 Q3).

Suppose we now have data from SRS, and we want an estimate of the population mean and 95% CI. We care about CIs because we want to know how uncertain our estimate is.

Definition 1.12 (**Standard Error (SE)**).
Standard error (SE) is an estimate of the standard deviation (of an estimator) based on a specific sample.

Example. For μ based on a SRS, the estimate is $\hat{\mu}$. We estimate $\operatorname{Var}(\tilde{\mu})$ with

$$\widehat{\operatorname{Var}}(\tilde{\mu}) = \left(1-\frac{n}{N}\right)\frac{\hat{\sigma}^2}{n},$$

and so $\operatorname{SE}(\hat{\mu}) = \sqrt{\widehat{\operatorname{Var}}(\tilde{\mu})} = \sqrt{\left(1-\frac{n}{N}\right)\frac{\hat{\sigma}^2}{n}}$, note that "hat" is used for σ^2.
For sampling, we will use Normal approximation to construct CIs, e.g. 95% CI for μ is

$$\hat{\mu} \pm 1.96\,\operatorname{SE}(\hat{\mu}).$$

Example. For τ (population total) in SRS, since $\tau = \sum_{i\in U} y_i = N\mu$, we have $\hat{\tau} = N\hat{\mu}$,

$$\operatorname{Var}(\tilde{\tau}) = \operatorname{Var}(N\tilde{\mu}) = N^2\operatorname{Var}(\tilde{\mu}) = N^2\left(1-\frac{n}{N}\right)\frac{\sigma^2}{n},$$

so $\widehat{\operatorname{Var}}(\tilde{\tau}) = N^2\widehat{\operatorname{Var}}(\tilde{\mu})$ and $\operatorname{SE}(\hat{\tau}) = N\cdot\operatorname{SE}(\hat{\mu})$. An approximate 95% CI for τ is

$$N\hat{\mu} \pm 1.96N\cdot\operatorname{SE}(\hat{\mu}).$$

Example (**Special Case**).

For proportion, suppose response is binary i.e.

$$y_i = \begin{cases} 1, & \text{answered yes,} \\ 0, & \text{answered no.} \end{cases}$$

The population mean of y_i's is called a **proportion**. Denote the population proportion as p. In this case,

$$\begin{aligned} \sigma^2 &= \frac{1}{N-1}\left(\sum_{i \in U} y_i^2 - N\mu^2\right) \\ &= \frac{1}{N-1}\left(Np - Np^2\right) \qquad (\text{since } y_i^2 = y_i \text{ and } \mu = p) \\ &= \frac{N}{N-1} p(1-p). \end{aligned}$$

Based on SRS,

$$\hat{\sigma}^2 = \frac{n}{n-1}\hat{p}(1-\hat{p}),$$

where $\hat{p}$ is the sample proportion. So a 95% CI for p is

$$\hat{p} \pm 1.96\,\mathrm{SE}(\hat{p}) = \hat{p} \pm 1.96\sqrt{\left(1 - \frac{n}{N}\right)\frac{\hat{p}(1-\hat{p})}{n-1}}.$$

Note. $\frac{n}{N}$ is the **sampling fraction** and $1 - \frac{n}{N}$ is the **fpc**. So if $n \approx N$, then $\mathrm{Var}(\tilde{\mu}) \to 0$, i.e. variability due to sampling $\to 0$. If $N \gg n$, then $\mathrm{fpc} = 1 - \frac{n}{N} \approx 1$.

Recall: How many Americans ($N = 342M$) vs. Canadians ($N = 41.5M$) for a sample? If $\hat{\sigma}^2$ is similar for US/Canada, then $\mathrm{SE}(\hat{\mu})$ is also very similar, even though N is much larger for US.

Example (**Sample Size Calculation**).

Suppose we have $\hat{\sigma}^2$ under SRS, how wide is my 95% CI if

(a) Sampled 5% of population, i.e. $\frac{n}{N} = 0.05$?

(b) Sampled 100 people?

Proof.

(a) We don't know since we need to know sample size n or population size N.

(b) We don't know since we need to know $\frac{n}{N}$ or N.

□

We need information on both n and N!

For the 95% CI for μ, the total CI width is

$$2 \cdot 1.96 \cdot \sqrt{\left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n}}.$$

So if we want the total width $< c$, this means

$$\begin{aligned}
\left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n} &< \left(\frac{c}{2 \cdot 1.96}\right)^2 \\
\frac{1}{n} - \frac{1}{N} &< \frac{1}{\hat{\sigma}^2} \cdot \left(\frac{c}{2 \cdot 1.96}\right)^2 \\
n &> \left[\frac{1}{\hat{\sigma}^2}\left(\frac{c}{2 \cdot 1.96}\right)^2 + \frac{1}{N}\right]^{-1}.
\end{aligned}$$

Don't forget to round n to integer!

Where to get $\hat{\sigma}^2$?

- Previous survey.
- In the special case of proportions,

$$\hat{\sigma}^2 = \frac{n}{n-1}\hat{p}(1-\hat{p}) \approx \hat{p}(1-\hat{p}) \le 0.5(1-0.5) = 0.25.$$

  This means that we can use 0.25 as a conservative upper bound for $\hat{\sigma}^2$.

Analyzing a SRS in R

The sampling designs in this class can be analyzed using the R package survey.

```
install.packages("survey")  # only need to run once
library(survey) # load the package

srs <- svydesign(ids=~1, data = dat, fpc = ~N)
# "ids=~1" tells R its a SRS, no clusters or strata
# "fpc" tells R which column is pop size to use for fpc
svymean(~sleep, srs)
# get estimate and SE for mean of variable "sleep"
```

```
mean(dat$sleep)
sqrt((1 - 1249/48415) * var(dat$sleep)/1249)
# can also manually calculate mean and SE like this
confint(svymean(~sleep, srs))
# get 95% CI for mean of "sleep"
```

Lecture 5, 2025/09/17

1.3 Questionnaire Design

When a survey respondent reads or hears a survey question, the following process usually occurs in their mind:

- Comprehension (interpreting the question)
- Retrieval (recall of information)
- Judgment (summarizing the information)
- Reporting (answering the question in the required format)

A well-designed questionnaire should make these steps as easy and efficient as possible for the respondent. If the process breaks down anywhere, answers collected will likely be unreliable (measurement error).

Comprehension problems:

- Poor or ambiguous wording.
- Too complex.
- Leading questions.
- False assumptions or inferences.

Retrieval problems:

- Lack of retrieval cues.
- Things are too hard to remember.
- Events happened too long ago.

Judgment and reporting problems:

- Response categories are too specific or don't cover all possibilities.
- Difficult to estimate based on question constraints.
- Stigma of giving certain responses.
- Deliberate misreporting.
  - People don't want to answer sensitive questions or reluctant to admit.

Response scale effects:

- People often don't want to pick options on the ends.
- Scale of -5 to $+5$ is interpreted differently than 0 to 10. Respondents tend to use higher ratings on the -5 to $+5$ scale.

Types of responses (respondents may be asked to answer in the following ways):

- Open-ended, with numerical response.
- Closed questions with ordered response scales.
- Closed questions with categorical response options.
  - Should be disjoint, and cover all possibilities.
  - Be specific.
  - Order matters: some respondents stop reading when they see an option that seems to describe them.

Strategies for sensitive questions:

- Give them a reasonable excuse to soften the behavior.
- Make it as an anonymous as possible.
- Randomized response.

Question testing:

- Survey questions often need multiple rounds of testing and feedback.

1.4 Ratio and Regression Estimation

Suppose we measure 2 variables/attributes, and we have the following:

- Frame: $U = \{1, 2, \ldots, N\}$.
- Values: $(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)$.
- Population quantities:
  (1) $\mu_X = \frac{1}{N}\sum_{i\in U} x_i$.
  (2) $\mu_Y = \frac{1}{N}\sum_{i\in U} y_i$.
  (3) Population correlation coefficient: $\rho = \frac{\sum_{i\in U}(x_i-\mu_X)(y_i-\mu_Y)}{(N-1)\sigma_x\sigma_y}$.
  (4) Population ratio of means: $\theta \overset{\text{def.}}{\equiv} \frac{\mu_Y}{\mu_X} = \frac{\frac{1}{N}\sum_{i\in U} y_i}{\frac{1}{N}\sum_{i\in U} x_i} = \frac{\tau_y}{\tau_x}$.

From an SRS of size n, we can estimate the following quantities:

(1) Sample correlation:
$$(r =)\, \hat{\rho} = \frac{\sum_{i\in S}(x_i - \hat{\mu}_x)(y_i - \hat{\mu}_y)}{(n-1)\hat{\sigma}_x\hat{\sigma}_y}.$$

(2) Sample ratio of means:
$$\hat{\theta} = \frac{\hat{\mu}_y}{\hat{\mu}_x}.$$

Question: Why consider θ?

(1) We want to estimate $\frac{\mu_Y}{\mu_X}$.

Example. Let

$$y_i = \text{\# of gators in tract } i,$$
$$x_i = \text{area of tract } i \text{ (in acres)}.$$

Then, θ is the average # of alligators per acre.

Example. Let

$$y_i = \text{\# served by employee } i \text{ in their shift,}$$
$$x_i = \text{length of that shift (in hours).}$$

Then, θ is the average # served per hour.

(2) When we know μ_X (population mean in this case), and we want to use it to get a more accurate estimate of μ_Y.

- If we don't know μ_X, we just estimate μ_Y with $\hat{\mu}_Y$.
- If we know μ_X, we can estimate μ_Y with $\hat{\mu}_Y \cdot \frac{\mu_X}{\hat{\mu}_X}$, in which we can think of $\frac{\mu_X}{\hat{\mu}_X}$ as an adjustment factor. Also,

$$\hat{\mu}_Y \cdot \frac{\mu_X}{\hat{\mu}_X} = \hat{\theta} \cdot \mu_X$$

which works well if x and y are highly correlated.

Properties of Estimator $\tilde{\theta} = \frac{\tilde{\mu}_Y}{\tilde{\mu}_X}$

Goal: Find $\mathbb{E}\left[\tilde{\theta}\right]$ and $\mathrm{Var}\left(\tilde{\theta}\right)$ under SRS. This is not trivial since $\mathbb{E}\left[\frac{W}{V}\right] \neq \frac{\mathbb{E}[W]}{\mathbb{E}[V]}$ in general.

Let's express $\tilde{\mu}_Y$ as the following:

$$\underbrace{\tilde{\mu}_Y}_{\text{r.v.}} = \underbrace{\mu_Y}_{\text{constant}} (1 + \underbrace{\varepsilon_Y}_{\text{r.v.}}).$$

Then,

$$\mu_Y = \mathbb{E}\left[\tilde{\mu}_Y\right] = \mu_Y(1 + \mathbb{E}\left[\varepsilon_Y\right]) \implies \mathbb{E}\left[\varepsilon_Y\right] = 0.$$

Similarly, we can express $\tilde{\mu}_X$ as

$$\tilde{\mu}_X = \mu_X(1 + \varepsilon_X), \quad \text{so } \mathbb{E}\left[\varepsilon_X\right] = 0.$$

We can write

$$\begin{aligned}
\tilde{\theta} = \frac{\tilde{\mu}_Y}{\tilde{\mu}_X} &= \frac{\mu_Y(1+\varepsilon_Y)}{\mu_X(1+\varepsilon_X)} \\
&= \frac{\mu_Y}{\mu_X}\cdot(1+\varepsilon_Y)(1-\varepsilon_X+\varepsilon_X^2-\varepsilon_X^3+\dots) && \text{by geometric series} \\
&= \frac{\mu_Y}{\mu_X}\left(1-\varepsilon_X+\varepsilon_Y-\varepsilon_X\varepsilon_Y+\varepsilon_X^2+\varepsilon_Y\varepsilon_X^2-\cdots\right) \\
&\approx \frac{\mu_Y}{\mu_X}(1-\varepsilon_X+\varepsilon_Y) && \text{by ignoring higher order terms.}
\end{aligned}$$

So, we have

$$\mathbb{E}\left[\tilde{\theta}\right] \approx \frac{\mu_Y}{\mu_X}(1-\mathbb{E}\left[\varepsilon_X\right]+\mathbb{E}\left[\varepsilon_Y\right]) = \frac{\mu_Y}{\mu_X}.$$

Thus, $\tilde{\theta}$ is approximately unbiased for θ. For the variance, we have

$$\begin{aligned}
\operatorname{Var}\left(\tilde{\theta}\right) &\approx \left(\frac{\mu_Y}{\mu_X}\right)^2 \operatorname{Var}\left(1-\varepsilon_X+\varepsilon_Y\right) \\
&= \left(\frac{\mu_Y}{\mu_X}\right)^2 \operatorname{Var}\left(1-\left(\frac{\tilde{\mu}_X}{\mu_X}-1\right)+\left(\frac{\tilde{\mu}_Y}{\mu_Y}-1\right)\right) \\
&= \left(\frac{\mu_Y}{\mu_X}\right)^2 \cdot \frac{1}{\mu_Y^2}\operatorname{Var}\left(\tilde{\mu}_Y-\mu_Y\cdot\frac{\tilde{\mu}_X}{\mu_X}\right) \\
&= \frac{1}{\mu_X^2}\operatorname{Var}\left(\tilde{\mu}_Y-\theta\tilde{\mu}_X\right).
\end{aligned}$$

Note that

$$\tilde{\mu}_Y-\theta\tilde{\mu}_X = \frac{1}{n}\sum_{i\in S} y_i - \theta\cdot\frac{1}{n}\sum_{i\in S} x_i = \frac{1}{n}\sum_{i\in S}(y_i-\theta x_i) = \frac{1}{n}\sum_{i\in S} r_i = \tilde{\mu}_r.$$

Hence, we know

$$\operatorname{Var}\left(\tilde{\mu}_r\right) = \left(1-\frac{n}{N}\right)\frac{\sigma_r^2}{n}, \quad \text{from SRS theory.}$$

So,

$$\operatorname{Var}\left(\tilde{\theta}\right) \approx \frac{1}{\mu_X^2}\left(1-\frac{n}{N}\right)\frac{\sigma_r^2}{n}$$

where

$$
\begin{aligned}
\sigma_r^2 &= \frac{1}{N-1}\sum_{i\in U}(r_i-\mu_r)^2 \\
&= \frac{1}{N-1}\sum_{i\in U}\left[(y_i-\theta x_i)-(\mu_Y-\theta\mu_X)\right]^2 \\
&= \frac{1}{N-1}\sum_{i\in U}\left[(y_i-\mu_Y)-\theta(x_i-\mu_X)\right]^2 \\
&= \frac{1}{N-1}\sum_{i\in U}\left[(y_i-\mu_Y)^2+\theta^2(x_i-\mu_X)^2-2\theta(y_i-\mu_Y)(x_i-\mu_X)\right] \\
&= \sigma_Y^2+\theta^2\sigma_X^2-2\theta\rho\sigma_X\sigma_Y.
\end{aligned}
$$

Therefore, we have

$$
\begin{aligned}
\mathbb{E}\left[\tilde{\theta}\right] &\approx \theta \\
\operatorname{Var}\left(\tilde{\theta}\right) &\approx \frac{1}{\mu_X^2}\left(1-\frac{n}{N}\right)\frac{1}{n}\left(\sigma_Y^2+\theta^2\sigma_X^2-2\theta\rho\sigma_X\sigma_Y\right).
\end{aligned}
$$

So to estimate θ based on SRS, we use

$$
\hat{\theta}=\frac{\hat{\mu}_Y}{\hat{\mu}_X}, \qquad \operatorname{SE}\left(\hat{\theta}\right)=\sqrt{\frac{1}{\hat{\mu}_X^2}\left(1-\frac{n}{N}\right)\frac{1}{n}\left(\hat{\sigma}_Y^2+\hat{\theta}^2\hat{\sigma}_X^2-2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y\right)}.
$$

And a 95% CI for θ is approximately

$$
\hat{\theta}\pm 1.96\,\operatorname{SE}\left(\hat{\theta}\right).
$$

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We can estimate $\operatorname{Var}\left(\tilde{\theta}\right)$ using

$$
\widehat{\operatorname{Var}}(\tilde{\theta})=\frac{1}{\hat{\mu}_X^2}\left(1-\frac{n}{N}\right)\frac{1}{n}\left(\hat{\sigma}_Y^2+\hat{\theta}^2\hat{\sigma}_X^2-2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y\right).
$$

Question: When will the estimation work well, i.e. when will $\operatorname{Var}\left(\tilde{\theta}\right)$ be small?

Answer:

- We want ρ to be large, i.e. $\rho\approx 1$.

Also note that

$$\sigma_Y^2 + \theta^2\sigma_X^2 - 2\theta\rho\sigma_X\sigma_Y = \sigma_r^2 = \frac{1}{N-1}\sum_{i\in U}(y_i - \theta x_i - (\mu_Y - \underbrace{\theta\mu_X}_{\mu_Y}))^2 = \frac{1}{N-1}\sum_{i\in U}(y_i - \theta x_i)^2.$$

- We want $y_i \approx \theta x_i$ for each unit i, i.e. y values are proportional to x values (can be described well as a straight line through the origin with slope θ).

Recall from last time that the CI for θ is $\hat{\theta} \pm 1.96\,\mathrm{SE}\left(\hat{\theta}\right)$. If we are interested in θ itself, we may also use $\tilde{\theta}$ to improve the accuracy of the estimate of μ_Y when μ_X is known. Since $\mu_Y = \theta\mu_X$, we can use $\tilde{\theta}\mu_X$ as an estimator for μ_Y. Then,

$$\begin{aligned}\mathbb{E}\left[\tilde{\theta}\mu_X\right] &= \mu_X\mathbb{E}\left[\tilde{\theta}\right] \approx \mu_X\frac{\mu_Y}{\mu_X} = \mu_Y,\\ \mathrm{Var}\left(\tilde{\theta}\mu_X\right) &= \mu_X^2\,\mathrm{Var}\left(\tilde{\theta}\right).\end{aligned}$$

Thus, we can construct the <u>ratio estimate of μ_Y</u> (if μ_X is known) as

$$\hat{\mu}_{\text{ratio}} = \hat{\theta}\mu_X = \frac{\hat{\mu}_Y}{\hat{\mu}_X}\mu_X.$$

The estimate of the variance is

$$\widehat{\mathrm{Var}}(\tilde{\mu}_{\text{ratio}}) = \mu_X^2\widehat{\mathrm{Var}}(\tilde{\theta}) = \frac{\mu_X^2}{\hat{\mu}_X^2}\left(1-\frac{n}{N}\right)\frac{1}{n}\left(\hat{\sigma}_Y^2 + \hat{\theta}^2\hat{\sigma}_X^2 - 2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y\right) \overset{\text{OR}}{\approx} \left(1-\frac{n}{N}\right)\frac{1}{n}\left(\hat{\sigma}_Y^2 + \hat{\theta}^2\hat{\sigma}_X^2 - 2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y\right).$$

Regression Estimation of the Mean

Consider the SRS with (x_i, y_i) measured and μ_X known, and the estimator

$$\tilde{\mu}_* = \tilde{\mu}_Y + b(\mu_X - \tilde{\mu}_X)$$

where b is a constant. Note that

- when $b = 0$, $\tilde{\mu}_* = \tilde{\mu}_Y$, the sample mean estimator for μ_Y (does not use any info about x).
- when $b = \tilde{\theta} = \frac{\tilde{\mu}_Y}{\tilde{\mu}_X}$, $\tilde{\mu}_* = \tilde{\mu}_Y + \frac{\tilde{\mu}_Y}{\tilde{\mu}_X}\mu_X - \tilde{\mu}_Y = \tilde{\theta}\mu_X = \hat{\mu}_{\text{ratio}}$.

Note. x and y need to have a linear relationship and not through the origin.

And for any value of b, we have

$$\begin{aligned}\mathbb{E}\left[\tilde{\mu}_*\right] &= \mathbb{E}\left[\tilde{\mu}_Y\right] + b(\mu_X - \mathbb{E}\left[\tilde{\mu}_X\right]) \\ &= \mu_Y + b(\mu_X - \mu_X) \\ &= \mu_Y, \text{ i.e. unbiased for any } b.\end{aligned}$$

So it makes sense to choose b to minimize $\operatorname{Var}(\tilde{\mu}_*)$. We have

$$\begin{aligned}\operatorname{Var}(\tilde{\mu}_*) &= \operatorname{Var}(\tilde{\mu}_Y + b(\mu_X - \tilde{\mu}_X)) \\ &= \operatorname{Var}(\tilde{\mu}_Y - b\tilde{\mu}_X).\end{aligned}$$

Note that this is the same form as for the ratio estimator except with $b = \theta$. So, we use the same approach to get

$$\operatorname{Var}(\tilde{\mu}_*) = \left(1 - \frac{n}{N}\right)\frac{1}{n}\left(\sigma_Y^2 + b^2\sigma_X^2 - 2b\rho\sigma_X\sigma_Y\right).$$

Let's find the optimal b to minimize this estimated variance, let

$$f(b) = \hat{\sigma}_Y^2 + b^2\hat{\sigma}_X^2 - 2b\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y$$

based on a sample. Then,

$$f'(b) = 2b\hat{\sigma}_X^2 - 2\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y \overset{\text{set}}{=} 0 \implies b = \frac{\hat{\rho}\hat{\sigma}_Y}{\hat{\sigma}_X}.$$

We can check this indeed minimizes $f(b)$ since $f''(b) = 2\hat{\sigma}_X^2 > 0$.

Let's call

$$\hat{\mu}_{\text{reg}} = \hat{\mu}_Y + \frac{\hat{\rho}\hat{\sigma}_Y}{\hat{\sigma}_X}(\mu_X - \hat{\mu}_X)$$

the regression estimate of μ_Y. The estimate of the variance is

$$\widehat{\operatorname{Var}}(\hat{\mu}_{\text{reg}}) = \left(1 - \frac{n}{N}\right)\frac{1}{n}\left(\hat{\sigma}_Y^2 + \frac{\hat{\rho}^2\hat{\sigma}_Y^2}{\hat{\sigma}_X^2}\hat{\sigma}_X^2 - 2\hat{\rho}\frac{\hat{\rho}\hat{\sigma}_Y}{\hat{\sigma}_X}\hat{\sigma}_X\hat{\sigma}_Y\right) = \left(1 - \frac{n}{N}\right)\frac{1}{n}(1 - \hat{\rho}^2)\hat{\sigma}_Y^2.$$

Remark (**Three ways to estimate** μ_Y).

$\hat{\mu}_Y$, $\hat{\mu}_{\text{ratio}}$, and $\hat{\mu}_{\text{reg}}$.

Question 1: When is $\hat{\mu}_{\text{ratio}}$ better than $\hat{\mu}_Y$? (i.e. lower estimated variance)

Answer: Recall that

$$\widehat{\text{Var}}(\tilde{\mu}_{\text{ratio}}) \approx \left(1-\frac{n}{N}\right)\frac{1}{n}\left(\hat{\sigma}_Y^2+\hat{\theta}^2\hat{\sigma}_X^2-2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y\right),$$
$$\widehat{\text{Var}}(\tilde{\mu}_Y) = \left(1-\frac{n}{N}\right)\frac{\hat{\sigma}_Y^2}{n}.$$

We want $\widehat{\text{Var}}(\tilde{\mu}_{\text{ratio}}) < \widehat{\text{Var}}(\tilde{\mu}_Y)$, i.e.

$$\hat{\sigma}_Y^2+\hat{\theta}^2\hat{\sigma}_X^2-2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y < \hat{\sigma}_Y^2$$
$$\hat{\theta}^2\hat{\sigma}_X^2-2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y < 0$$
$$\hat{\rho} > \frac{\hat{\theta}\hat{\sigma}_X}{2\hat{\sigma}_Y} = \frac{1}{2}\frac{\hat{\mu}_Y\hat{\sigma}_X}{\hat{\mu}_X\hat{\sigma}_Y}.$$

So if x, y have similar means/variances, then $\hat{\rho} > \frac{1}{2}$ is needed for ratio estimation to be better.

Question 2: When is $\hat{\mu}_{\text{reg}}$ better than $\hat{\mu}_Y$?

Answer: Recall that $\widehat{\text{Var}}(\tilde{\mu}_{\text{reg}}) = \left(1-\frac{n}{N}\right)\frac{1}{n}(1-\hat{\rho}^2)\hat{\sigma}_Y^2$. We want $\widehat{\text{Var}}(\tilde{\mu}_{\text{reg}}) < \widehat{\text{Var}}(\tilde{\mu}_Y)$, i.e.

$$1-\hat{\rho}^2 < 1 \implies \hat{\rho}^2 > 0 \implies \hat{\rho} \neq 0.$$

As long as $\hat{\rho} \neq 0$, regression estimation is essentially always better than the sample mean.

Question 3: When is $\hat{\mu}_{\text{reg}}$ better than $\hat{\mu}_{\text{ratio}}$?

Answer: See A2 Q3.

Ratio and Regression Estimation in R

```
library(survey)
# Set up the things we know
srs <- svydesign(id=~1, fpc=~N, data=dat) # SRS design
mu.x <- data.frame(Farms = 2087759/3144) # known mean of x

ratio <- svyratio(~Output, ~Farms, design=srs)
predict(ratio, total = mu.x)
regression <- svyglm(Output~Farms, design=srs)
predict(regression, newdata = mu.x)

svymean(~Output, srs)
```

Lecture 8, 2025/09/29

1.5 Stratified Sampling

Divide the population of size N into H strata (subpopulations) such that

$$N = \sum_{h=1}^{H} N_h$$

where N_h is the population size of stratum h. We have

Stratum	Units	Measurements/Values
1	$\{1, 2, \ldots, N_1\}$	$\{y_{11}, y_{12}, \ldots, y_{1N_1}\}$
2	$\{1, 2, \ldots, N_2\}$	$\{y_{21}, y_{22}, \ldots, y_{2N_2}\}$
$\vdots$	$\vdots$	$\vdots$
H	$\{1, 2, \ldots, N_H\}$	$\{y_{H1}, y_{H2}, \ldots, y_{HN_H}\}$

Remark. Stratified sampling works well when units in a stratum are similar among themselves.

Assume we take a SRS of size n_h from N_h in each stratum $h = 1, 2, \ldots, H$ independently.

Population quantities:

- μ: overall population mean (over all N units).
- μ_h: population mean of stratum h.
- σ_h^2: population variance of stratum h.

Sample quantities:

- $\hat{\mu}_h$: sample mean of stratum h.
- $\hat{\sigma}_h^2$: sample variance of stratum h.

The key quantity to estimate is μ. We can write μ as:

$$\mu = \frac{1}{N} \sum_{h=1}^{H} \sum_{j=1}^{N_h} y_{hj} = \frac{1}{N} \sum_{h=1}^{H} N_h \mu_h = \sum_{h=1}^{H} \frac{N_h}{N} \mu_h.$$

where $\sum_{j=1}^{N_h} y_{hj}$ is the population total of stratum h, $\frac{N_h}{N}$ is the fraction of population that belongs to stratum h. Let $W_h = \frac{N_h}{N}$ be the "stratum weights". Then,

$$\mu = \sum_{h=1}^{H} W_h \mu_h.$$

So a natural estimate for μ is

$$\hat{\mu}_S = \sum_{h=1}^{H} W_h \hat{\mu}_h.$$

Question: How many possible samples are there for stratified sampling vs. overall SRS?

Example. $(N = 80, n = 20)$ vs. $(N_1 = N_2 = N_3 = N_4 = 20$ and $n_1 = n_2 = n_3 = n_4 = 5)$.

- Overall SRS: $\binom{80}{20}$.
- Stratified SRS: $\binom{20}{5}^4$ where $\binom{20}{5}$ is the # of ways to choose 5 from 20 in each stratum.

Note that $\binom{80}{20} > \binom{20}{5}^4$. By stratifying, we ensure that we get exactly 5 from each stratum achieving a balanced sample which is not guaranteed with an overall SRS.

Mean and Variance of $\tilde{\mu}_S$

Note that

$$\mathbb{E}\left[\tilde{\mu}_S\right] = \mathbb{E}\left[\sum_{h=1}^{H} W_h \tilde{\mu}_h\right] = \sum_{h=1}^{H} W_h \mathbb{E}\left[\tilde{\mu}_h\right] = \sum_{h=1}^{H} W_h \mu_h = \mu$$

since we have SRS in each stratum. Thus, $\tilde{\mu}_S$ is unbiased for μ.

Also,

$$\begin{aligned}
\operatorname{Var}\left(\tilde{\mu}_S\right) = \operatorname{Var}\left(\sum_{h=1}^{H} W_h \tilde{\mu}_h\right) &= \sum_{h=1}^{H} W_h^2 \operatorname{Var}\left(\tilde{\mu}_h\right) \quad \text{(independent strata)} \\
&= \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\sigma_h^2}{n_h}.
\end{aligned}$$

Allocation

For a fixed $n = \sum_{h=1}^{H} n_h$ (total sample size), how to choose $n_1, n_2, \ldots, n_H$?

Approach 1: Use the same sampling fraction in each stratum. This is called proportional allocation:

$$\frac{n}{N} = \frac{n_h}{N_h} \quad \forall h,$$

where $\frac{n}{N}$ is the overall sampling fraction and $\frac{n_h}{N_h}$ is the sampling fraction in stratum h.

Example. Let $N_1 = 100, N_2 = 200$, say we can sample a total of $n = 30$. Then,

$$\frac{n}{N} = \frac{30}{300} = 0.1 \implies n_1 = 10, n_2 = 20.$$

That is, $n_h = \frac{n}{N} N_h = W_h n \propto W_h$.

Under proportional allocation, we have

$$\begin{aligned}
\operatorname{Var}\left(\tilde{\mu}_{S,\text{prop}}\right) &= \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\sigma_h^2}{n_h} \\
&= \sum_{h=1}^{H} W_h \frac{N_h}{N} \frac{1}{n_h} \left(1 - \frac{n}{N}\right) \sigma_h^2 \\
&= \sum_{h=1}^{H} W_h \left(1 - \frac{n}{N}\right) \frac{\sigma_h^2}{n}.
\end{aligned}$$

Question: How does this compare to $\operatorname{Var}(\tilde{\mu})$ for an overall SRS (size n from N)?

Note that

$$\frac{\operatorname{Var}\left(\tilde{\mu}_{S,\text{prop}}\right)}{\operatorname{Var}(\tilde{\mu})} = \frac{\sum_{h=1}^{H} W_h \left(1 - \frac{n}{N}\right) \frac{\sigma_h^2}{n}}{\left(1 - \frac{n}{N}\right) \frac{\sigma^2}{n}} = \frac{\sum_{h=1}^{H} W_h \sigma_h^2}{\sigma^2}.$$

Since $\sum_{h=1}^{H} W_h = 1$, stratified sampling is good if σ_h^2 (variance within strata) is small relative to σ^2 (overall population variance).

Decompose the population variance σ^2 in stratified sampling:

$$\begin{aligned}
\sigma^2 &= \frac{1}{N-1}\sum_{h=1}^{H}\sum_{j=1}^{N_h}(y_{hj}-\mu)^2 \qquad \text{definition of pop. variance in stratified sampling}\\
&= \frac{1}{N-1}\sum_{h=1}^{H}\sum_{j=1}^{N_h}\left[(y_{hj}-\mu_h)+(\mu_h-\mu)\right]^2\\
&= \frac{1}{N-1}\sum_{h=1}^{H}\sum_{j=1}^{N_h}\left[(y_{hj}-\mu_h)^2+(\mu_h-\mu)^2+2(y_{hj}-\mu_h)(\mu_h-\mu)\right]\\
&= \frac{1}{N-1}\left[\sum_{h=1}^{H}\sigma_h^2(N_h-1)+\sum_{h=1}^{H}N_h(\mu_h-\mu)^2+2\sum_{h=1}^{H}\left((\mu_h-\mu)\underbrace{\sum_{j=1}^{N_h}(y_{hj}-\mu_h)}_{=N_h\mu_h-N_h\mu_h=0}\right)\right]\\
&= \sum_{j=1}^{N_h}\frac{N_h-1}{N-1}\sigma_h^2+\sum_{h=1}^{H}\frac{N_h}{N-1}(\mu_h-\mu)^2.
\end{aligned}$$

So if N_h and N are large, we have $\frac{N_h-1}{N-1}\approx\frac{N_h}{N}=W_h$. Thus,

$$\sigma^2\approx\sum_{h=1}^{H}W_h\sigma_h^2+\sum_{h=1}^{H}W_h(\mu_h-\mu)^2.$$

Note. $\sigma_h^2=\dfrac{1}{N_h-1}\displaystyle\sum_{j=1}^{N_h}(y_{hj}-\mu_h)^2$ is the population variance of stratum h and $N_h\mu_h$ is the population total of stratum h.

Recall that allocation is choosing $n_1,n_2,\ldots,n_H$ for a fixed $n=\sum_{h=1}^{H}n_h$. The first approach is proportional allocation: $n_h\propto W_h$ and last time we found that

$$\begin{aligned}
\frac{\operatorname{Var}\left(\tilde{\mu}_{S,\text{prop}}\right)}{\operatorname{Var}\left(\tilde{\mu}\right)} = \frac{\sum_{h=1}^{H}W_h\sigma_h^2}{\sigma^2} &\approx \frac{\sum_{h=1}^{H}W_h\sigma_h^2}{\sum_{h=1}^{H}W_h\sigma_h^2+\sum_{h=1}^{H}W_h(\mu_h-\mu)^2}\geq 0\\
&\leq 1.
\end{aligned}$$

Last time: Stratum works well if σ_h^2 are small relative to σ^2, turns out it's equivalent conceptually to if μ_h are very different compared to μ.

Approach 2: Optimal allocation (i.e. minimize variance) to choose $n_1, n_2, \dots, n_H$.

If we have some knowledge of σ_h^2, we can use that to set $n_1, \dots, n_H$ to minimize $\operatorname{Var}(\tilde{\mu}_S)$ for a fixed $n = \sum_{h=1}^H n_h$.

Problem: Minimize $\operatorname{Var}(\tilde{\mu}_S) = \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\sigma_h^2}{n_h}$ as a function of $n_1, n_2, \dots, n_H$ with constraint $\sum_{h=1}^{H} n_h = n$. This can be solved with Lagrange multipliers. Setup:

$$L(n_1, n_2, \dots, n_H) = \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\sigma_h^2}{n_h} + \lambda \left(\underbrace{\sum_{h=1}^{H} n_h - n}_{\text{constraint} = 0} \right).$$

Set $\frac{\partial L}{\partial n_h} = 0$ for each $h = 1, \dots, H$, and $\frac{\partial L}{\partial \lambda} = 0$ to get (PS4 Q2):

$$n_h = \frac{W_h \sigma_h}{\sum_{h=1}^H W_h \sigma_h} \cdot n \propto W_h \sigma_h$$

and the denominator $\sum_{h=1}^H W_h \sigma_h$ does not depend on h.

Optimal allocation says we sample more from stratum with:

(1) Large N_h (large W_h).

(2) Large variance (σ_h high).

Example.

Stratum	W_h	$\hat{\sigma}_h$	$W_h \hat{\sigma}_h$
1	0.25	4	1
2	0.25	8	2
3	0.5	4	2

Note that we plug in our guess/estimate $\hat{\sigma}_h$ for σ_h in practice.

Say we can do $n = 100$ surveys in total.

(1) Proportional allocation: $n_1 = 25$, $n_2 = 25$, $n_3 = 50$.

(2) Optimal allocation: $n_1 = \frac{1}{5} \cdot 100 = 20$, $n_2 = \frac{2}{5} \cdot 100 = 40$, $n_3 = \frac{2}{5} \cdot 100 = 40$. And

$$\sum_{h=1}^{H} W_h \hat{\sigma}_h = \sum_{h=1}^{3} W_h \hat{\sigma}_h = 1 + 2 + 2 = 5.$$

Question: When is proportional allocation also optimal?

Answer: When σ_h are all the same, i.e. $\sigma_1 = \sigma_2 = \ldots = \sigma_H$.

Analyzing a Stratified Sample in R

To estimate the number of births in a country in a year, a researcher compiled a list of 158 hospitals in the country.

- The researcher divided hospitals into strata based on their size: 42 small, 99 medium, and 17 large.
- The researcher visited an SRS of 4 small, 5 medium, and 6 large hospitals, all sampled independently from their strata.

```
library(survey)
dat <- read.csv("births.csv")
boxplot(dat$births~dat$strata) # boxplots by strata
### Set up svydesign:
strat <- svydesign(id=~1, fpc=~N, strata=~strata, data=dat)
# the "strata" tells R where the strata are coded
# what's different about how "N" is specified?
svymean(~births, strat)
svytotal(~births, strat)
confint(svytotal(~births, strat))
# Can also look at totals by stratum:
svyby(~births, ~strata, strat, svytotal)
```

Stratified Sampling in Practice

To set up a stratified sampling design, we need:

- # of units in each stratum: $N_1, N_2, \dots, N_H$.
- A list of units for each stratum.
- Each unit should belong to exactly one stratum.
- (Optional) For optimal allocation, good guesses of σ_h^2; otherwise, use proportional allocation.

Why stratify?

- If it makes sense to divide the population into strata that are internally similar.
- To achieve desired representativeness from each stratum.
- Can get lower variance estimate using the same total sample size (compared to SRS on entire population), if the stratum means μ_h are very different from the overall mean μ. Equivalently, this means σ_h^2 within strata are smaller than the overall variance σ^2.

Why not stratify?

- It's too much work.

What if we are not sure how different the strata will be?

- Doesn't matter, if strata are large and n_h are reasonably large (≥ 30 or so), then FPCs ≈ 1.
- Use proportional allocation.
- Under these conditions, stratified variance will not be worse than taking an overall SRS.

Poststratification

Suppose we know

- # of units in each stratum: $N_1, N_2, \dots, N_H$.
- Each unit belongs to exactly one stratum.

However, if our frame does not tell us which units belong to which stratum, we cannot take a stratified sample. But we can analyze the data as if it came from a stratified sampling design. This is called **poststratification**.

Steps to do poststratification:

(1) Decide strata beforehand just like for stratified sampling.

(2) Take a SRS from the entire population, and ask each sampled unit which stratum it belongs to.

(3) Use survey data to determine $n_1, n_2, \dots, n_H$.

(4) Analyze the data as if it came from a stratified sampling design, the post-stratified estimate of μ is

$$\hat{\mu}_{\text{post}} = \sum_{h=1}^{H} W_h \hat{\mu}_h.$$

(5) The poststratified variance can be approximated using the stratified variance formula for proportional allocation:

$$\widehat{\text{Var}}(\tilde{\mu}_{\text{post}}) = \sum_{h=1}^{H} W_h \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}_h^2}{n}.$$

Example (**Poststratification Example**).

Suppose you take a survey of Vancouver, want to stratify by ethnicity. Instead, take SRS of $n = 1000$ since we don't have a list of ethnicities.

Stratum	W_h	n_h
White	0.48	$550 \to \hat{\mu}_1$
Chinese	0.27	$200 \to \hat{\mu}_2$
Other Asian	0.19	$150 \to \hat{\mu}_3$
Other	0.06	$100 \to \hat{\mu}_4$

Note that w_h are from the census and n_h represents the number after the survey. Then,

$$\hat{\mu}_{\text{post}} = 0.48\hat{\mu}_1 + 0.27\hat{\mu}_2 + 0.19\hat{\mu}_3 + 0.06\hat{\mu}_4.$$

μ_{post} adjusts estimate to each stratum's actual share of population.

Lecture 10, 2025/10/06

1.6 Cluster Sampling

Divide the population into clusters.

Sampling design: we sample clusters, then take measurements on some or all units in the sampled clusters.

Terms:

- PSU: primary sampling unit (i.e. cluster).
- SSU: secondary sampling unit (i.e. unit within cluster).

Example.

(1) Imagine we are doing a neighbourhood survey. We may have the following clusters.
  - PSU: block of houses or apartment buildings.
  - SSU: one residental unit.

(2) Cows on a farm. We may have:
  - PSU: farm.
  - SSU: cow.

Question: Why do we use cluster sampling?

Answer:

- For convenience, e.g. main cost is to access clusters.
- It is easy to sample/measure units in the same cluster.

In practice, units within the same cluster tend to be more similar among themselves compared to the population as a whole.

Notation: Assume the population is made up of A clusters, each with B units (equal-sized clusters), so the population size is $N = AB$.

First, take SRS of size a from A clusters, then take SRS of size b from B units in each sampled cluster. So, the total sample size is $n = ab$.

Case 1: $b = B$, i.e. measure all units of the sampled clusters.

Suppose we have $\mu_1, \mu_2, \dots, \mu_A$ are cluster means in the population. Then, the population mean is

$$\mu = \frac{1}{A} \sum_{a=1}^{A} \mu_a \qquad \text{since equal-sized clusters assumed.}$$

After sampling, we observe values μ_i where indices i are a SRS of size a from $\{1, 2, \dots, A\}$. From which, we estimate μ using

$$\hat{\mu}_{\text{clu}} = \frac{1}{a} \sum_{i \in S} \mu_i, \quad \text{with } \operatorname{Var}(\tilde{\mu}_{\text{clu}}) = \left(1 - \frac{a}{A}\right) \frac{\sigma_A^2}{a}$$

where $\sigma_A^2 = \frac{1}{A-1} \sum_{a=1}^{A} (\mu_a - \mu)^2$ is the population variance of cluster means.

Example. There are 20 cows on 4 farms. We have the following data on weights of cows (in 1000's of lbs):

Farm	Individual weights					μ_a
1	1.3	1.3	1.6	1.4	1.4	1.4
2	1.2	1.4	1.3	1.1	1.5	1.3
3	1.4	1.6	1.5	1.5	1.5	1.5
4	1.0	1.1	1.3	0.9	1.2	1.1

Assume that $N = 20$ is the whole population. Then, the population mean of 20 cows is

$$\mu = \frac{1.4 + 1.3 + 1.5 + 1.1}{4} = 1.325$$

The population variance of 20 cows is

$$\sigma^2 = \frac{1}{20 - 1} \left[(1.3 - 1.325)^2 + (1.3 - 1.325)^2 + \cdots + (1.2 - 1.325)^2 \right] = 0.03776.$$

The population variance of 4 farms means is

$$\sigma_A^2 = \frac{1}{4 - 1} \left[(1.4 - 1.325)^2 + (1.3 - 1.325)^2 + (1.5 - 1.325)^2 + (1.1 - 1.325)^2 \right] = 0.02917.$$

Consider 2 sampling designs:

(1) SRS of 10 cows from 20 cows.

(2) SRS of 2 farms from 4 farms, weigh all cows in the sampled farms.

How do their estimation variances compare?

(1) $\left(1-\frac{10}{20}\right)\frac{0.03776}{10}=0.0019.$
(2) $\left(1-\frac{2}{4}\right)\frac{0.02917}{2}=0.0073.$
Note that the second variance is almost 4 times larger since 5 cows from the same farm gives less information than 5 cows from random farms.

Note. Cluster sampling is not as efficient as SRS. But, it's more convenient and cheaper in practice.

In general, say n is the total sample size. Then,

(1) SRS of n from N has variance $\left(1-\dfrac{n}{N}\right)\dfrac{\sigma^2}{n}$.

(2) SRS of $a=\frac{n}{B}$ clusters and observe all B units in each sampled cluster has variance $\left(1-\dfrac{a}{A}\right)\dfrac{\sigma_A^2}{a}$.

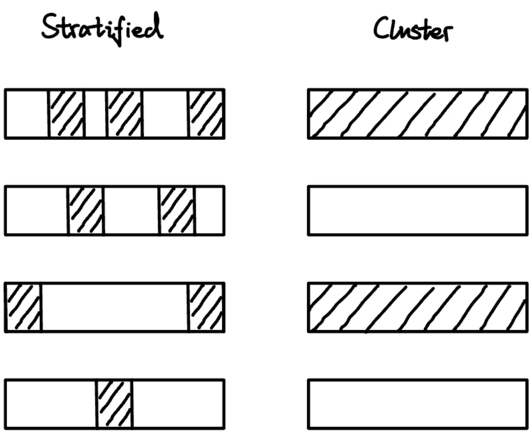
Since n is the same,

$$\frac{n}{N}=\frac{aB}{AB}=\frac{a}{A}.$$

Then,

$$\frac{\operatorname{Var}\left(\tilde{\mu}_{\text{clu}}\right)}{\operatorname{Var}\left(\tilde{\mu}\right)}=\frac{\left(1-\frac{a}{A}\right)\frac{\sigma_A^2}{a}}{\left(1-\frac{n}{N}\right)\frac{\sigma^2}{n}}=\frac{B\sigma_A^2}{\sigma^2}.$$

In practice, units in the same cluster are similar, so σ_A^2 tends to be not much smaller than σ^2. So usually, cluster sampling has larger variance for the same n.



- Stratified sampling works well when units in strata are similar.
- Cluster sampling works poorly when units in clusters are similar.

Stratified sampling vs. cluster sampling:

- Stratified sampling works well if units within a stratum are similar among themselves. Samples from every stratum.
- Cluster sampling works poorly (i.e. high estimation variance) if units within a cluster are similar among themselves. Only get samples from some clusters.

Example. Suppose we have the following data.

Cluster	Unit Measurements	Cluster Mean
1	$y_{11}, y_{12}, \cdots, y_{1B}$	μ_1
2	$y_{21}, y_{22}, \cdots, y_{2B}$	μ_2
3	$y_{31}, y_{32}, \cdots, y_{3B}$	μ_3
$\vdots$	$\vdots$	$\vdots$
A	$y_{A1}, y_{A2}, \cdots, y_{AB}$	μ_A

Recall that the population variance of cluster means is

$$\sigma_A^2 = \frac{1}{A-1}\sum_{a=1}^{A}(\mu_a - \mu)^2,$$

which is different from the population variance of all units

$$\sigma^2 = \frac{1}{AB-1}\sum_{a=1}^{A}\sum_{b=1}^{B}(y_{ab} - \mu)^2.$$

Claim. If units within clusters are similar, then σ_A^2 will not be much smaller than σ^2.

Proof. Define

$$\sigma_B^2 = \underbrace{\frac{1}{A}\sum_{a=1}^{A}\underbrace{\left[\frac{1}{B-1}\sum_{b=1}^{B}(y_{ab} - \mu_a)^2\right]}_{\text{pop. variance of cluster } a}}_{\text{average within-cluster variance}}.$$

Then, we can decompose σ^2 as follows:

$$\begin{aligned}
\sigma^2 &= \frac{1}{AB-1}\sum_{a=1}^{A}\sum_{b=1}^{B}(y_{ab}-\mu_a+\mu_a-\mu)^2 \\
&= \frac{1}{AB-1}\sum_{a=1}^{A}\sum_{b=1}^{B}\left[(y_{ab}-\mu_a)^2+(\mu_a-\mu)^2+2(y_{ab}-\mu_a)(\mu_a-\mu)\right] \\
&= \frac{1}{AB-1}\left[\sum_{a=1}^{A}\sum_{b=1}^{B}(y_{ab}-\mu_a)^2+B\sum_{a=1}^{A}(\mu_a-\mu)^2+2\sum_{a=1}^{A}(\mu_a-\mu)\underbrace{\sum_{b=1}^{B}(y_{ab}-\mu_a)}_{=B\mu_a-B\mu_a=0}\right]
\end{aligned}$$

Note that $B\mu_a$ is the cluster total of cluster a.

$$= \frac{1}{AB-1}\left[\sigma_B^2\cdot A(B-1)+\sigma_A^2\cdot B(A-1)\right].$$

So if A and B are large, then

$$\sigma^2 \approx \sigma_A^2+\sigma_B^2$$

i.e. population variance $\approx$ variance of cluster means + average within-cluster variance. In practice, σ_B^2 tends to be small, hence σ_A^2 will not be that small relative to σ^2. □

Case 1: SRS of a out of A clusters measure all B units in each sampled cluster.

$$\hat{\mu}_{\text{clu}} = \frac{1}{a}\sum_{i\in S}\hat{\mu}_i, \quad \operatorname{Var}(\tilde{\mu}_{\text{clu}}) = \left(1-\frac{a}{A}\right)\frac{\sigma_A^2}{a}.$$

Recall that

$$\frac{\operatorname{Var}(\tilde{\mu}_{\text{clu}})}{\operatorname{Var}(\tilde{\mu})} = \frac{B\sigma_A^2}{\sigma^2} \quad \text{often } > 1.$$

Case 2: Take SRS of a clusters out of A clusters, then take SRS of size b units out of B units in each sampled cluster. Overall sample size is $n = ab$. Now, instead of observing μ_i for sampled clusters i, we only get $\hat{\mu}_i$ from the sample mean of the b units out of B units in cluster i.

Then,

$$\hat{\mu}_{\text{clu},2} = \frac{1}{a}\sum_{i\in S}\hat{\mu}_i,$$
$$\text{Var}\left(\tilde{\mu}_{\text{clu},2}\right) = \left(1-\frac{a}{A}\right)\frac{\hat{\sigma}_A^2}{a} + \left(1-\frac{b}{B}\right)\frac{\hat{\sigma}_B^2}{Ab}.$$

The second term $\left(1-\frac{b}{B}\right)\frac{\hat{\sigma}_B^2}{ab}$ is the extra variance from having to estimate μ_i.

- If $b = B$, reduces to Case 1.
- In real situations, cluster sizes will not necessarily be equal (R survey package has formulas for that).

Analyzing a Cluster Sample in R

The survey package contains a California school performance dataset that we'll use as our example. The data is built-in so we can load it like this:

```
library(survey)
data(api)
```

There are two hypothetical cluster samples included:

- apiclus1: Take SRS of school districts and measure performance at all schools in sampled districts (case 1)
- apiclus2: Take SRS of school districts, then take SRS of schools within the sampled districts (case 2)

There are many variables in this dataset, let's focus on "college graduation percentage" (col.grad).

```
## dnum = district number (cluster)
## snum = school number (unit within cluster)
## Case 1
## tapply computes a function on a vector, by subgroup,
## e.g. cluster means:
tapply(apiclus1$col.grad, apiclus1$dnum, mean)
## fpc = number of districts
clus1 <- svydesign(id=~dnum, data=apiclus1, fpc=~fpc)
svymean(~col.grad, clus1)
```

```
# Case 2
## fpc1 = number of districts
## fpc2 = number of schools in that district
clus2 <- svydesign(id=~dnum+snum, data=apiclus2, fpc=~fpc1+fpc2)
svymean(~col.grad, clus2)
```

Lecture 12, 2025/10/20

1.7 Non-Response and Non-Representative Survey Data (Not Tested)

To wrap up sampling designs, we look at two common practical situations.

Definition 1.13 (**Non-Response**).
A sampled unit does not participate in the survey at all, or does not answer some of the survey questions.

Definition 1.14 (**Non-Representative Samples**).
The sample does not reflect the target population.

Example (**Non-Representative Samples Example**).
With mass internet and social media, it is often easy to obtain large numbers of survey respondents for little cost - but is such data useful at all?

So far, we've assumed all sampled units respond completely to the survey. In practice, this is often not true. There are two types of non-response:

Definition 1.15 (**Unit Non-Response**).
When a unit selected in the sample does not participate in the survey.

Definition 1.16 (**Item Non-Response**).
When a unit selected in the sample answers some, but not all, items of the survey (e.g. person misses some questions or refuses to answer some sensitive questions)

Unit non-response in a SRS

Let's briefly talk about a statistical framework for unit non-response.

As before, suppose that we have an ideal frame for the target population $\{1, 2, \ldots, N\}$ with population mean μ, and we take a SRS of size n. Now, think of the population as being divided into two groups:

- Those who would respond to the survey.
- Those who would not respond to the survey.

We use the following notation:

- N_R: size of stratum for those who would respond.
- N_M: size of stratum for those who would not respond.
- μ_R: mean for those who would respond.
- μ_M: mean for those who would not respond.

Then,

$$\begin{aligned} \mu &= W_R \mu_R + W_M \mu_M \\ &= \frac{N_R}{N} \mu_R + \frac{N_M}{N} \mu_M. \end{aligned}$$

Question: Can we actually post-stratify respondents/non-respondents based on an overall SRS?

Answer: No, since we don't know who would respond until after we take the sample, i.e. we don't know $\hat{\mu}_M$. We also don't know the proportions W_R and W_M. We have

$$\hat{\mu}_{\text{post}} = \underbrace{W_R}_{?} \hat{\mu}_R + \underbrace{W_M}_{?} \underbrace{\hat{\mu}_M}_{?}.$$

Question: What happens if we only use the respondents in the sample to estimate μ?

If we use $\tilde{\mu}_R$ as an estimator for μ:

$$\mathbb{E}[\tilde{\mu}_R] = \mu_R,$$

but we actually want to estimate μ. The **non-response bias** is

$$\begin{aligned}\mathbb{E}\left[\tilde{\mu}_R\right] - \mu = \mu_R - \mu = \mu_R - \left(\frac{N_R}{N}\mu_R + \frac{N_M}{N}\mu_M\right) \\ = \frac{N_R + N_M}{N}\mu_R - \frac{N_R}{N}\mu_R - \frac{N_M}{N}\mu_M \\ = \frac{N_M}{N}(\mu_R - \mu_M)\end{aligned}$$

where $\frac{N_M}{N}$ is the fraction of non-respondents and $\mu_R - \mu_M$ is the difference between the mean of respondents and the mean of non-respondents.

Note. This bias is 0 when $\mu_R = \mu_M$.

Non-response bias becomes larger when:

- The difference between μ_R and μ_M gets larger.
- The proportion of non-respondents in the population gets larger.

Ways to correct for non-response

(1) Follow-up with initial non-respondents, hopefully getting some of them to respond, and thus estimate μ_M. Plug into post-stratification formula:

$$\hat{\mu}_{\text{post}} = \frac{n_R}{n}\hat{\mu}_R + \frac{n_M}{n}\hat{\mu}_M$$

where $\frac{n_R}{n}$ is the fraction of initial sample that responded.

(2) Use what we have and post-stratify with demographic data to compensate for non-response, e.g. by asking demographic questions on the survey, and then use population stratum weights from census data.

Example (**Previous Poststratification Example**).

Suppose you take a survey of Vancouver, want to stratify by ethnicity. Instead, take SRS of $n = 1000$ since we don't have a list of ethnicities.

Stratum	W_h	n_h (ideal)	respondents
White	0.48	550	$400 \to \hat{\mu}_1$
Chinese	0.27	200	$100 \to \hat{\mu}_2$
Other Asian	0.19	150	$60 \to \hat{\mu}_3$
Other	0.06	100	$70 \to \hat{\mu}_4$

Note that the n_h column is the ideal case where everyone responds. The last column is the actual number of respondents from each stratum. Then,

$$\hat{\mu}_{\text{post}} = 0.48\hat{\mu}_1 + 0.27\hat{\mu}_2 + 0.19\hat{\mu}_3 + 0.06\hat{\mu}_4.$$

This assumes respondents in each stratum are representative of the whole stratum.

Key assumption for non-representative samples: respondents in each stratum are representative of the whole stratum.

Non-Representative Samples

Probability-based sampling designs are the preferred way to obtain reliable survey data that can be generalized to a target population. In practice, it is quite difficult to obtain truly representative samples (e.g. due to non-response). It is also relatively costly (time and money) to run such a survey, and total sample size will often be limited.

With the arrival of mass internet and social media, it is easy to obtain a much larger number of respondents for a fraction of the cost. However, such data will usually be non-representative of the population.

Maybe we can post-stratify to correct for the non-representative sample, similar to non-response correction.

Response rate adjustment

Conceptually, we might pretend the sample is a huge SRS with a huge difference in response rates among the different demographic groups. Then, one way to compensate for the non-representative sample is through post-stratification by demographic groups. We post-stratify by all combinations of the demographic variables.

By dividing up the population so finely, the goal is that units in each stratum are similar among themselves, so that respondents in each stratum are representative of the whole stratum.

However, even with a large sample size, some post-strata may have very few respondents. So we fit a model to fill in the gaps.

Model-based post-stratification approach

- Fit a logistic regression model (e.g. where 1 = 'vote Democratic', 0 = 'vote Republican') on the demographic variables.
- Use the model to predict the proportion voting Democratic in each post-stratum $\hat{p}_h$.
- Get the actual stratum weights W_h in the population from census data.
- Use post-stratification to get the adjusted estimate

$$\hat{p}_{\text{post}} = \sum_{h=1}^{H} W_h \hat{p}_h.$$

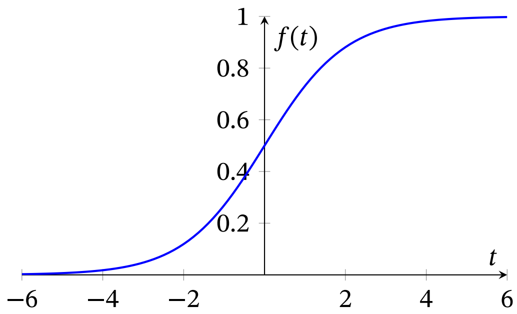
- On the Xbox dataset, this solved the problem of empty post-strata and produced an adjusted estimate that was only 0.6% off from the true election result.

Logistic Regression Crash Course

The **logistic function** is

$$f(t) = \frac{1}{1 + e^{-t}}.$$

It looks like this:



This is often used to model probabilities for binary responses (i.e. yes/no). We interpret the value of

$f(t)$ as the proportion of "yes".

Example. Suppose we use the logistic function to model the probability of Donald Trump's current job approval for each voter affiliation: Democrats, Republicans, and Independents, with the model

$$\frac{1}{1+e^{0.5+3.5d-3r}}$$

where $d = 1$ if Democrat (0 otherwise); $r = 1$ if Republican (0 otherwise).

Then, the proportion approving of Donald Trump's current job performance based on the model is

- Democrat: $\frac{1}{1+e^{0.5+3.5}} = 0.02$.
- Republican: $\frac{1}{1+e^{0.5-3}} = 0.92$.
- Independent: $\frac{1}{1+e^{0.5}} = 0.38$.

In practice, we use R's `glm()` function to fit the model. For example,

```
dat <- read.csv("trump-approval.csv")
table(dat$gender, dat$age, dat$party_affiliation)
lmodel <- glm(approve ~ factor(gender)
 + factor(party_affiliation) + factor(age_group),
   data = dat, family = "binomial")
summary(lmodel)
```

2 Experimental Design

Lecture 13, 2025/10/22

2.1 Fundamentals of Experimental Design

Survey Sampling

- Used for descriptive study: we obtain information about variables of interest via measurement on units from the population.
- The measurement y_i is treated as fixed (non-random) once unit i is included in the sample.

Experimental Design

- Used for experimental study: we deliberately change input(s) to assess their effect on an outcome.
- The outcome (or response) is treated as random: the outcomes of units assigned the same setting of the input variables have a probability distribution. In STAT 332, we assume these outcomes follow a Normal distribution.

Example. We are interested in studying whether coffee or tea, and chocolate or maple donuts, work better at keeping students awake in class. Let's recruit 200 students for this experiment: 100 from a statistics course and 100 from a math course. We will give each student one drink (coffee or tea) and one donut (chocolate or maple), with an equal number of math and statistics receiving each possible combination. The students then attend a lecture from their respective courses, with the lecture continuing until all students fall asleep. The time it takes each student to fall asleep is recorded, along with which drink and donut they had.

Definition 2.1 (**Factor**).
A **factor** is an input variable that is controlled or set by the experimenter.

Example. The type of drink; the type of donut.

Definition 2.2 (**Factor Levels**).

The **factor levels** are the different values assigned to a factor.

Example. Coffee or tea (2 levels); chocolate or maple (2 levels).

Definition 2.3 (**Treatment**).

A **treatment** is a combination of the factor levels.

Example. Coffee with a chocolate donut (the example has $2 \times 2 = 4$ treatments).

Definition 2.4 (**Experimental Unit**).

An **experimental unit** is an object or individual on which treatment is applied.

Example. A student.

Definition 2.5 (**Response**).

The **response** is the measured outcome.

Example. The time (say, in minutes) until a student falls asleep in lecture.

Note. In STAT 332, we assume the response is continuous and follows a Normal distribution.

Three Fundamental Principles

(1) **Replication**: applying each treatment to more than one unit.

(2) **Randomization**: assigning treatments to units randomly.

(3) **Blocking**: dividing the experimental units into groups with similar characteristics.

Model for a Single Mean/Treatment

$$Y_i = \mu + \epsilon_i, \quad i = 1, 2, \dots, n,$$

where

- Y_i is the response of unit i (random variable).
- μ is the overall mean.
- $\epsilon_i \sim \mathrm{N}(0, \sigma^2)$ accounts for variability/randomness in the response and are independent between units.

We make the following remarks for this model:

- Observe data $y_1, y_2, \dots, y_n$.
- Estimate parameters using the least squares:

$$\min_{\mu} \sum_{i=1}^{n} (y_i - \mu)^2 \implies \hat{\mu} = \bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i.$$

- Using estimates of model parameters, get estimates of σ^2:

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (y_i - \hat{\mu})^2 = \frac{1}{n-1} \sum_{i=1}^{n} (y_i - \bar{y})^2.$$

- Corresponding estimators $\tilde{\mu}$, $\tilde{\sigma}^2$ have distributions:

$$\tilde{\mu} \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{\tilde{\sigma}^2(n-1)}{\sigma^2} \sim \chi^2_{n-1}.$$

  Then,

$$\frac{\tilde{\mu} - \mu}{\sigma/\sqrt{n}} \sim \mathrm{N}(0,1), \quad \text{and} \quad \frac{\tilde{\mu} - \mu}{\tilde{\sigma}/\sqrt{n}} \sim t_{n-1}.$$

- Note that

$$\mathrm{SE}\,(\hat{\mu}) = \frac{\hat{\sigma}}{\sqrt{n}}.$$

  So, the $(1-\alpha)\%$ CI for μ is

$$\hat{\mu} \pm c\,\mathrm{SE}\,(\hat{\mu})$$

  where c is the value where $\mathbb{P}(|T| \leq c) = 1 - \alpha$ for $T \sim t_{n-1}$.

Lecture 14, 2025/10/27

- Hypothesis tests: H_0: $\mu = \mu_0$ vs. H_A: $\mu \neq \mu_0$. If H_0 is true, then

$$\frac{\tilde{\mu} - \mu_0}{\tilde{\sigma}/\sqrt{n}} \sim t_{n-1}.$$

  So, we can calculate the test statistic

$$t_{\text{obs}} = \frac{\hat{\mu} - \mu_0}{\hat{\sigma}/\sqrt{n}}$$

  and we reject H_0 at level α if $|t_{\text{obs}}| > c$. Next, we can calculate the p-value as

$$\text{p-value} = \mathbb{P}(|T| \geq |t_{\text{obs}}|) = 2\mathbb{P}(T \geq |t_{\text{obs}}|) \quad \text{for } T \sim t_{n-1}.$$

Example. Suppose we have $n = 20$ experimental units, $\hat{\mu} = 2$, and $\hat{\sigma} = 1.5$. To get a 95% CI for μ, we have $\alpha = 0.05$ and

$$2 \pm c\frac{1.5}{\sqrt{20}}, \quad \mathbb{P}(|T| \leq c) = 0.95 \text{ for } T \sim t_{19}.$$

We can use R:

```
> qt(0.975, 19)
[1] 2.093024
```

So, $c = 2.093$. Now, we test $H_0 : \mu = 1$ vs. $H_A : \mu \neq 1$. Then,

$$t_{\text{obs}} = \frac{2-1}{1.5/\sqrt{20}} = 2.981.$$

At $\alpha = 0.05$, we have $c = 2.093$. Since $|t_{\text{obs}}| > c$, we reject H_0 at level 0.05. The p-value is

$$\text{p-value} = 2\mathbb{P}(T \geq 2.981) \quad \text{for } T \sim t_{19}.$$

Using R:

```
> pt(2.981, 19)
[1] 0.9961611
```

So, the p-value is $2 \times (1 - 0.9961611) = 0.0077$.

2.2 Two Treatment Models

The model is

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}, \quad i = 1, 2, \quad j = 1, 2, \dots, n_i$$

where

- n_1 is the # of units in treatment 1; n_2 is the # of units in treatment 2, and $n = n_1 + n_2$.
- μ is the overall mean.
- τ_1 and τ_2 are the treatment effects.
- $\epsilon_{ij} \sim \mathrm{N}(0, \sigma^2)$ are all independent, assume the same σ^2 for both treatments.

Note that the mean in treatment i is $\mu + \tau_i$ and the mean in treatment 2 is $\mu + \tau_2$.

- We denote $\cdot$ average over that subscript. Here, $\bar{Y}_{i\cdot}$ is the average of units in treatment i:

$$\bar{Y}_{1\cdot} = \frac{1}{n_1}\sum_{j=1}^{n_1} Y_{1j}, \quad \bar{Y}_{2\cdot} = \frac{1}{n_2}\sum_{j=1}^{n_2} Y_{2j}.$$

The average of all units is

$$\bar{Y}_{\cdot\cdot} = \frac{1}{n_1 + n_2}\sum_{i=1}^{2}\sum_{j=1}^{n_i} Y_{ij}.$$

Then,

$$\begin{aligned}
\mathbb{E}\left[\bar{Y}_{1\cdot}\right] &= \frac{1}{n_1}\sum_{j=1}^{n_1}\mathbb{E}\left[Y_{1j}\right] = \frac{1}{n_1}\sum_{j=1}^{n_1}(\mu + \tau_1) = \mu + \tau_1, \\
\mathbb{E}\left[\bar{Y}_{2\cdot}\right] &= \mu + \tau_2.
\end{aligned}$$

The overall mean should be represented by μ, so we want $\mu = \mathbb{E}\left[\bar{Y}_{\cdot\cdot}\right]$. Note that

$$\begin{aligned}
\mu = \mathbb{E}\left[\bar{Y}_{\cdot\cdot}\right] &= \frac{1}{n_1 + n_2}\left[n_1(\mu + \tau_1) + n_2(\mu + \tau_2)\right] \\
&= \mu + \frac{n_1\tau_1 + n_2\tau_2}{n_1 + n_2}.
\end{aligned}$$

So, $n_1\tau_1 + n_2\tau_2 = 0$ is a required constraint.

Observe data:

- Treatment 1: $y_{11}, y_{12}, \dots, y_{1n_1}$.
- Treatment 2: $y_{21}, y_{22}, \dots, y_{2n_2}$.

Estimate parameters using least squares:

$$\min_{\mu,\tau_1,\tau_2} \sum_{i=1}^{2} \sum_{j=1}^{n_i} (y_{ij} - \mu - \tau_i)^2 \quad \text{subject to } n_1\tau_1 + n_2\tau_2 = 0.$$

Using Lagrange multipliers, we have

$$\hat{\mu} = \bar{y}_{..}, \quad \hat{\tau}_1 = \bar{y}_{1.} - \bar{y}_{..}, \quad \hat{\tau}_2 = \bar{y}_{2.} - \bar{y}_{..}.$$

Then, we estimate σ^2 as

$$\hat{\sigma}^2 = \frac{1}{n_1 + n_2 - 2} \sum_{i=1}^{2} \sum_{j=1}^{n_i} (y_{ij} - \hat{\mu} - \hat{\tau}_i)^2.$$

We can show that

$$\hat{\sigma}^2 = \frac{(n_1 - 1)\hat{\sigma}_1^2 + (n_2 - 1)\hat{\sigma}_2^2}{n_1 + n_2 - 2}$$

which is in terms of sample variances of each treatment.

We focus on the "difference between treatment effects", i.e. $\tau_1 - \tau_2$. This is estimated by

$$\hat{\tau}_1 - \hat{\tau}_2,$$

and $\tilde{\tau}_1 - \tilde{\tau}_2 \sim \mathrm{N}\left(\tau_1 - \tau_2, \sigma^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)\right)$. So,

$$\frac{\tilde{\sigma}^2\,(n_1 + n_2 - 2)}{\sigma^2} \sim \chi^2_{n_1+n_2-2}.$$

Thus,

$$\frac{(\tilde{\tau}_1 - \tilde{\tau}_2) - (\tau_1 - \tau_2)}{\tilde{\sigma}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}.$$

Also,

$$\mathrm{SE}\,(\tilde{\tau}_1 - \tilde{\tau}_2) = \hat{\sigma}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$

So, a $(1-\alpha)\%$ CI for $\tau_1 - \tau_2$ is

$$\hat{\tau}_1 - \hat{\tau}_2 \pm c\,\mathrm{SE}\,(\hat{\tau}_1 - \hat{\tau}_2)$$

where c is the value where $\mathbb{P}(|T| \leq c) = 1 - \alpha$ for $T \sim t_{n_1+n_2-2}$.

For hypothesis tests, we test $H_0 : \tau_1 - \tau_2 = 0$ (no difference in treatment effects) vs. $H_A : \tau_1 - \tau_2 \neq 0$. The test statistic is

$$t_{\text{obs}} = \frac{\hat{\tau}_1 - \hat{\tau}_2}{\hat{\sigma}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

look up values from $t_{n_1+n_2-2}$ distribution.

Lecture 15, 2025/10/29

Example. See slides.

2.3 Two Treatment Models with Blocks (Single Replicate)

The new model is

$$Y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}, \quad i = 1, 2, \quad j = 1, 2, \dots, b$$

where b is the # of blocks, and β_j is the block effect.

The constraints are $\tau_1 + \tau_2 = 0$ and $\sum_{j=1}^{b} \beta_j = 0$. Assume that the treatment has the same effect on each block.

Estimate parameters:

$$\min_{\mu,\tau_i,\beta_j} \sum_{i=1}^{2} \sum_{j=1}^{b} (y_{ij} - \mu - \tau_i - \beta_j)^2 \quad \text{subject to} \begin{cases} \tau_1 + \tau_2 = 0, \\ \sum_{j=1}^{b} \beta_j = 0. \end{cases}$$

Then, we have

$$\hat{\mu} = \bar{y}_{..}, \quad \hat{\tau}_i = \bar{y}_{i.} - \bar{y}_{..}, \quad \hat{\beta}_j = \bar{y}_{.j} - \bar{y}_{..}.$$

The estimate of σ^2 is

$$\begin{aligned}\hat{\sigma}^2 &= \frac{1}{b-1}\sum_{i=1}^{2}\sum_{j=1}^{b}(y_{ij}-\hat{\mu}-\hat{\tau}_i-\hat{\beta}_j)^2 \\ &= \frac{1}{2}\times \text{“ sample variance of the differences } y_{1j}-y_{2j}\text{”}.\end{aligned}$$

The parameter of interest is $\tau_1 - \tau_2$, which can be estimated by $\hat{\tau}_1 - \hat{\tau}_2$.

Inference:

$$\tilde{\tau}_1 - \tilde{\tau}_2 \sim \mathrm{N}\left(\tau_1 - \tau_2, \frac{2\sigma^2}{b}\right) \implies \frac{\tilde{\tau}_1 - \tilde{\tau}_2}{\sigma\sqrt{\frac{2}{b}}} \sim \mathrm{N}(0,1).$$

Then,

$$\frac{(\tilde{\tau}_1 - \tilde{\tau}_2) - (\tau_1 - \tau_2)}{\tilde{\sigma}\sqrt{\frac{2}{b}}} \sim t_{b-1}$$

and

$$\mathrm{SE}\,(\hat{\tau}_1 - \hat{\tau}_2) = \hat{\sigma}\sqrt{\frac{2}{b}}.$$

Hypothesis test: $H_0 : \tau_1 = \tau_2$ vs. $H_A : \tau_1 \neq \tau_2$. The test statistic is

$$t_{\text{obs}} = \frac{\hat{\tau}_1 - \hat{\tau}_2 - 0}{\hat{\sigma}\sqrt{\frac{2}{b}}} \quad \text{compared to } t_{b-1}.$$

Basic Distribution Result for Experimental Design

- If $Y \sim \mathrm{N}(\mu, \sigma^2)$, then $\frac{Y-\mu}{\sigma} \sim \mathrm{N}(0,1)$ and $\left(\frac{Y-\mu}{\sigma}\right)^2 \sim \chi^2_1$.
- Linear combinations of independent Normals are also Normal. Suppose $Y_i \sim \mathrm{N}(\mu_i, \sigma_i^2)$ for all $i = 1, 2, \ldots, n$ are all independent. Then, for constants a_i,

$$\sum_{i=1}^{n} a_i Y_i \sim \mathrm{N}\left(\sum_{i=1}^{n} a_i\mu_i, \sum_{i=1}^{n} a_i^2\sigma_i^2\right).$$

- If $U \sim \chi^2_m$ and $V \sim \chi^2_n$ are independent, then $U + V \sim \chi^2_{m+n}$.
- For linear models, suppose $\hat{y}_i$ is a fitted value for y_i and the residuals follow $\mathrm{N}(0, \sigma^2)$ and are independent. Then,

$$\frac{\sum_{i=1}^{n}(Y_i - \tilde{Y}_i)}{\sigma^2} \sim \chi^2_{\text{df}}$$

where df = degrees of freedom = sample size − # of free model parameters. And,

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\text{df}}$$

is the estimate of σ^2. Also,

$$\frac{\tilde{\sigma}^2 \cdot \text{df}}{\sigma^2} \sim \chi^2_{\text{df}}$$

because

$$\frac{\frac{\sum_{i=1}^{n}(Y_i - \tilde{Y}_i)^2}{\text{df}} \cdot \text{df}}{\sigma^2} = \frac{\sum_{i=1}^{n}(Y_i - \tilde{Y}_i)^2}{\sigma^2} \sim \chi^2_{\text{df}}.$$

- t-distribution is $T = \frac{Z}{\sqrt{\frac{U}{k}}}$ where $Z \sim \text{N}(0,1)$ and $U \sim \chi^2_k$ are independent. Then, $T \sim t_k$.

Two Treatment Model with Replicates

The model is

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

where df $= \underbrace{n_1 + n_2}_{\text{sample size}} -2$ since we estimate μ, τ_1, and τ_2 with $n_1\tau_1 + n_2\tau_2 = 0$.

Question: What is the distribution of $\bar{Y}_{1\cdot}$and $\bar{Y}_{2\cdot}$?

Answer: Normal distribution since they are linear combinations of independent Normals.

Note. The distribution of $\tilde{\tau}_1 - \tilde{\tau}_2 = \bar{Y}_{1\cdot} - \bar{Y}_{2\cdot}$ is also Normal.

Thus,

$$\frac{\frac{\tilde{\tau}_1 - \tilde{\tau}_2 - (\tau_1 - \tau_2)}{\hat{\sigma}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}}{\sqrt{\frac{\tilde{\sigma}^2(n_1+n_2-2)}{\sigma^2} \cdot \frac{1}{n_1+n_2-2}}} = \frac{\tilde{\tau}_1 - \tilde{\tau}_2}{\hat{\sigma}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}.$$

Note that

$$\frac{\tilde{\tau}_1 - \tilde{\tau}_2 - (\tau_1 - \tau_2)}{\hat{\sigma}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim \text{N}(0,1) \quad \text{and} \quad \frac{\tilde{\sigma}^2(n_1 + n_2 - 2)}{\sigma^2} \sim \chi^2_{n_1+n_2-2}$$

Two Treatment Model with Blocks

The model is

$$Y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}$$

and there are $b + 3$ parameters: μ, τ_1, τ_2, and $\beta_1, \beta_2, \ldots, \beta_b$. There are 2 constraints: $n_1\tau_1 + n_2\tau_2 = 0$

and $\sum_{j=1}^{b} \beta_j = 0$. Then, df $= 2b - (b + 3 - 2) = b - 1$.

Lecture 16, 2025/11/03

2.4 Completely Randomized Design (CRD) with Multiple (more than two) Treatments

Consider balanced design (same # of replicates in each treatment) with t treatments, each with r replicates, i.e. $n = tr$. The model is

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}, \quad i = 1, 2, \ldots, t, \quad j = 1, 2, \ldots, r$$

where

- μ is the overall mean.
- τ_i is the treatment effect for treatment i.
- $\epsilon_{ij} \sim \mathrm{N}(0, \sigma^2)$ are independent, all treatments have the same σ^2.
- $\mu + \tau_i$ is the mean response in treatment i.

We want $\mu = \mathbb{E}\left[\bar{Y}_{\cdot\cdot}\right]$. This leads to the constraint $\sum_{i=1}^{t} \tau_i = 0$, where τ_i is the effect of treatment i relative to the overall mean.

Estimation:

$$\min_{\mu, \tau_i} \sum_{i=1}^{t} \sum_{j=1}^{r} (y_{ij} - \mu - \tau_i)^2 \quad \text{subject to } \sum_{i=1}^{t} \tau_i = 0.$$

This gives

$$\hat{\mu} = \bar{y}_{\cdot\cdot}, \quad \hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}.$$

Let's look at the 2-treatment model with replicates:

Example. Estimated difference between treatment i and j is $\hat{\tau}_i - \hat{\tau}_j = \bar{y}_{i\cdot} - \bar{y}_{j\cdot}$. Here,

$$\text{df} = \text{sample size} - \text{\# of free model parameters} = tr - (\underbrace{1}_{\mu} + \underbrace{t}_{\tau_i} - \underbrace{1}_{\text{constraint}}) = t(r-1).$$

The estimate of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{t(r-1)} \sum_{i=1}^{t} \sum_{j=1}^{r} (y_{ij} - \hat{\mu} - \hat{\tau}_i)^2.$$

Inference:

$$\bar{Y}_{i\cdot} \sim \mathrm{N}\left(\mu + \tau_i, \frac{\sigma^2}{r}\right) \quad \text{and} \quad \frac{\tilde{\sigma}^2(t(r-1))}{\sigma^2} \sim \chi^2_{t(r-1)}.$$

With t treatments, there is many possible combinations of treatment effects that we might consider testing for significance.

Example. Suppose we have $t = 4$ treatments as follows.

Treatments	Drink	Donut	Effect
1	Coffee	Chocolate	τ_1
2	Coffee	Maple	τ_2
3	Tea	Chocolate	τ_3
4	Tea	Maple	τ_4

We might be interested in testing the following hypotheses:

(1) $H_0 : \tau_1 - \tau_2 = 0$. Interpretation: when drink is coffee, there is no difference between donuts.

(2) $H_0 : \tau_1 - \tau_3 = 0$. Interpretation: when donut is chocolate, there is no difference between drinks.

(3) $H_0 : \frac{\tau_1+\tau_2}{2} - \frac{\tau_3+\tau_4}{2} = 0$. Interpretation: there is no difference between coffee and tea overall.

(4) $H_0 : (\tau_1 - \tau_3) - (\tau_2 - \tau_4) = 0$. Interpretation: the effect of coffee vs. tea is the same for either donut.

These are all examples of **contrasts**.

Definition 2.6 (Contrast).

A **contrast** θ is a linear combination of treatment effects

$$\theta = \sum_{i=1}^{t} a_i \tau_i \text{ s.t. } \sum_{i=1}^{t} a_i = 0$$

for constants a_i.

Example. In the previous example, we have the following contrasts:

Contrast	a_1	a_2	a_3	a_4
(1)	1	-1	0	0
(2)	1	0	-1	0
(3)	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$
(4)	1	-1	-1	1

Estimate of a contrast θ is

$$\hat{\theta} = \sum_{i=1}^{t} a_i \hat{\tau}_i = \sum_{i=1}^{t} a_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) = \sum_{i=1}^{t} a_i \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} \underbrace{\sum_{i=1}^{t} a_i}_{=0} = \sum_{i=1}^{t} a_i \bar{y}_{i\cdot}\,.$$

The estimator $\tilde{\theta} = \sum_{i=1}^{t} a_i \bar{Y}_{i\cdot}$ is Normal:

$$\mathbb{E}\left[\tilde{\theta}\right] = \sum_{i=1}^{t} a_i \mathbb{E}\left[\bar{Y}_{i\cdot}\right] = \sum_{i=1}^{t} a_i(\mu + \tau_i) = \sum_{i=1}^{t} a_i \tau_i = \theta \implies \text{unbiased.}$$

Also,

$$\operatorname{Var}\left(\tilde{\theta}\right) = \sum_{i=1}^{t} a_i^2 \operatorname{Var}\left(\bar{Y}_{i\cdot}\right) = \frac{\sigma^2}{r} \sum_{i=1}^{t} a_i^2.$$

So,

$$\tilde{\theta} \sim \mathrm{N}\left(\theta, \frac{\sigma^2}{r} \sum_{i=1}^{t} a_i^2\right) \implies \frac{\tilde{\theta} - \theta}{\tilde{\sigma}\sqrt{\frac{\sum_{i=1}^{t} a_i^2}{r}}} \sim t_{t(r-1)}.$$

Example. Suppose there are 50 students per treatment. We test $H_0 : \tau_1 - \tau_2 = 0$. Then,

$$t_{\text{obs}} = \frac{\hat{\tau}_1 - \hat{\tau}_2}{\hat{\sigma}\sqrt{\frac{1^2 + (-1)^2}{50}}} = \frac{\hat{\tau}_1 - \hat{\tau}_2}{\hat{\sigma}\sqrt{\frac{1}{50} + \frac{1}{50}}}.$$

This is the same t_{obs} as 2-treatment model with replicates! But, compare t_{obs} to t_{196} (not t_{98} since we used all 4 treatments to estimate parameters).

For $H_0 : \frac{\tau_1+\tau_2}{2} - \frac{\tau_3+\tau_4}{2} = 0$, we have

$$t_{\text{obs}} = \frac{\frac{\hat{\tau}_1+\hat{\tau}_2}{2} - \frac{\hat{\tau}_3+\hat{\tau}_4}{2}}{\hat{\sigma}\sqrt{\frac{\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^2+\left(-\frac{1}{2}\right)^2+\left(-\frac{1}{2}\right)^2}{50}}} = \frac{\frac{\hat{\tau}_1+\hat{\tau}_2}{2} - \frac{\hat{\tau}_3+\hat{\tau}_4}{2}}{\hat{\sigma}\sqrt{\frac{1}{100}+\frac{1}{100}}}$$

compared to t_{196}.

Lecture 17, 2025/11/05

Contrasts lead to t-tests for linear combinations of treatment effects.

Example. $H_0 : \tau_1 - \frac{\tau_2+\tau_3}{2} = 0$.

Since there are t treatments, we can also ask: "are all treatment effects the same?" This leads to $H_0 : \tau_1 = \tau_2 = \cdots = \tau_t = 0$ vs. $H_A : \tau_i \neq 0$ for at least one treatment i. If H_0 is true, then $Y_{ij} = \mu + \epsilon_{ij}$, i.e. all treatments have the same μ.

We use analysis of variance (ANOVA) to test this H_0.

Fact 1: If $U \sim \chi^2_m$, then $\mathbb{E}[U] = m$. So,

$$\frac{\tilde{\sigma}^2 t(r-1)}{\sigma^2} \sim \chi^2_{t(r-1)} \implies \mathbb{E}\left[\frac{\tilde{\sigma}^2 t(r-1)}{\sigma^2}\right] = t(r-1) \implies \mathbb{E}\left[\tilde{\sigma}^2\right] = \sigma^2 \quad \text{(unbiased)}.$$

Fact 2: F distribution is $\frac{U/m}{V/n} \sim F_{m,n}$ where $U \sim \chi^2_m$ and $V \sim \chi^2_n$ are independent.

If H_0 is true, then $\bar{Y}_{i.} \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{r}\right)$. We can think of $\bar{Y}_{i.} = \mu + \epsilon'_i$ where $\epsilon'_i \sim \mathrm{N}\left(0, \frac{\sigma^2}{r}\right)$, $i = 1, 2, \dots, t$, which looks like a single mean/treatment model.

So we can estimate $\frac{\sigma^2}{r}$ using sample variance of $\bar{y}_{i.}$:

$$\frac{1}{t-1}\sum_{i=1}^{t}(\bar{y}_{i.} - \bar{y}_{..})^2.$$

So,

$$\tilde{\sigma}_t^2 = \frac{r}{t-1}\sum_{i=1}^{t}(\bar{Y}_{i.} - \bar{Y}_{..})^2 \quad \text{is unbiased for } \sigma^2 \text{ if } H_0 \text{ is true.}$$

Also,

$$\frac{\tilde{\sigma}_t^2(t-1)/r}{\sigma^2/r} \sim \chi^2_{t-1}.$$

And we have 2 ways to estimate σ^2, where $\tilde{\sigma}_t^2$ only works if H_0 is true.

If H_0 is true,

$$\frac{\frac{\tilde{\sigma}_t^2(t-1)}{\sigma^2}\cdot\frac{1}{t-1}}{\frac{\tilde{\sigma}_t^2 t(r-1)}{\sigma^2}\cdot\frac{1}{t(r-1)}} = \frac{\tilde{\sigma}_t^2}{\tilde{\sigma}^2} \sim F_{t-1,t(r-1)}.$$

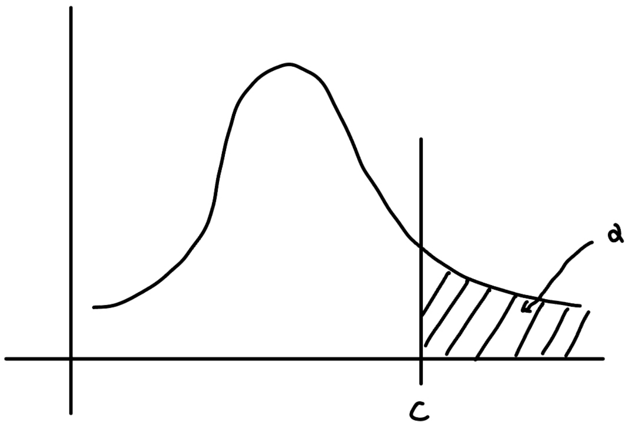
If H_0 is false, then $\tilde{\sigma}_t^2$ is not unbiased for σ^2. In fact,

$$\mathbb{E}\left[\tilde{\sigma}_t^2\right] = \sigma^2 + \frac{r}{t-1}\sum_{i=1}^{t}\tau_i^2 \quad \text{(see PS7).}$$

Thus,

$$F_{\text{obs}} = \frac{\hat{\sigma}_t^2}{\hat{\sigma}^2}$$

is evidence against H_0 if it is large.



That is, we reject H_0 if $F_{\text{obs}} > c$ where c is such that $\mathbb{P}(F_{t-1,t(r-1)} \geq c) = \alpha$ at level α.

Note. F-test is always one-sided (on the right tail) whereas t-test can be two-sided.

Sum of Squares Decomposition (SS)

- Variance of the entire sample (not σ^2):

$$\frac{1}{tr-1}\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{\cdot\cdot})^2.$$

- Estimate of residual variance:

$$\hat{\sigma}^2 = \frac{1}{t(r-1)}\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{i\cdot})^2.$$

- From F-test,

$$\hat{\sigma}_t^2 = \frac{r}{t-1}\sum_{i=1}^{t}(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})^2 = r \times \text{“variance of } t \text{ treatment means”}.$$

Then,

$$\begin{aligned}\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{\cdot\cdot})^2 &= \sum_{i=1}^{t}\sum_{j=1}^{r}\left[(y_{ij}-\bar{y}_{i\cdot})+(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})\right]^2 \\ &= \sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{i\cdot})^2 + \sum_{i=1}^{t}\sum_{j=1}^{r}(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})^2 + 2\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{i\cdot})(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot}).\end{aligned}$$

Note that

$$2\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{i\cdot})(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot}) = 2\sum_{i=1}^{t}(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})\underbrace{\sum_{j=1}^{r}(y_{ij}-\bar{y}_{i\cdot})}_{r\bar{y}_{i\cdot}-r\bar{y}_{i\cdot}=0} = 0.$$

Thus,

$$\underbrace{\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{\cdot\cdot})^2}_{\text{Total SS}} = \underbrace{\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{i\cdot})^2}_{\text{Error/Residual SS (RSS)}} + \underbrace{\sum_{i=1}^{t}(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})^2}_{\text{Treatment SS}}.$$

So, Total SS = Treatment SS + RSS.

We have the following ANOVA table:

Source	SS	df	Mean Square $= \frac{\text{SS}}{\text{df}}$	F stat
Treatment	$r\sum_{i=1}^{t}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$	$t-1$	$\hat{\sigma}_t^2$	$F_{\text{obs}} = \frac{\hat{\sigma}_t^2}{\hat{\sigma}^2}$
Residuals	$\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij} - \bar{y}_{i\cdot})^2$	$t(r-1)$	$\hat{\sigma}^2$	
Total	$\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij} - \bar{y}_{\cdot\cdot})^2$	$tr-1$		

Lecture 18, 2025/11/10

2.5 CRD with Unbalanced Replicates

Unbalanced: potentially different # of replicates in each treatment.

Let r_i be the # of replicates in treatment i, the model is

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}, \quad i = 1, 2, \dots, t, \quad j = 1, 2, \dots, r_i$$

where τ_i is the effect of treatment i and $\epsilon_{ij} \sim \mathrm{N}(0, \sigma^2)$ are independent.

We want μ to represent the overall mean:

$$\mu = \mathbb{E}\left[\bar{Y}_{\cdot\cdot}\right] = \mathbb{E}\left[\frac{1}{\sum_{i=1}^{t} r_i}\sum_{i=1}^{t}\sum_{j=1}^{r_i} Y_{ij}\right] = \mathbb{E}\left[\frac{\sum_{i=1}^{t} r_i \bar{Y}_{i\cdot}}{\sum_{i=1}^{t} r_i}\right] = \frac{\sum_{i=1}^{t} r_i(\mu + \tau_i)}{\sum_{i=1}^{t} r_i} = \mu + \frac{\sum_{i=1}^{t} r_i \tau_i}{\sum_{i=1}^{t} r_i}.$$

Note. $\sum_{i=1}^{t} r_i$ is the total sample size and

$$\bar{Y}_{i\cdot} = \frac{1}{r_i}\sum_{j=1}^{r_i} Y_{ij} \sim \mathrm{N}\left(\mu + \tau_i, \frac{\sigma^2}{r_i}\right).$$

We want $\mathbb{E}\left[\bar{Y}_{\cdot\cdot}\right] = \mu$, which leads to the constraint

$$\sum_{i=1}^{t} r_i \tau_i = 0.$$

Estimation:

$$\min_{\mu, \tau_i} \sum_{i=1}^{t}\sum_{j=1}^{r_i}(y_{ij} - \mu - \tau_i)^2 \quad \text{subject to } \sum_{i=1}^{t} r_i \tau_i = 0.$$

This gives

$$\hat{\mu} = \bar{y}_{..}, \quad \hat{\tau}_i = \bar{y}_{i.} - \bar{y}_{..}.$$

The degree of freedom is

$$\mathrm{df} = \underbrace{\sum_{i=1}^{t} r_i}_{\text{sample size}} - (\underbrace{1}_{\mu} + \underbrace{t}_{\tau_i} - \underbrace{1}_{\text{constraint}}) = \sum_{i=1}^{t} r_i - t.$$

The estimate of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{\sum_{i=1}^{t} r_i - t} \sum_{i=1}^{t} \sum_{j=1}^{r_i} (\underbrace{y_{ij} - \hat{\mu} - \hat{\tau}_i}_{y_{ij} - \bar{y}_{i.}})^2.$$

Contrasts: $\bar{Y}_{i.} \sim \mathrm{N}\left(\mu + \tau_i, \frac{\sigma^2}{r_i}\right)$. If $\theta = \sum_{i=1}^{t} a_i \tau_i$ is a contrast, we estimate it by

$$\hat{\theta} = \sum_{i=1}^{t} a_i \hat{\tau}_i = \sum_{i=1}^{t} a_i \bar{y}_{i.}.$$

And,

$$\tilde{\theta} \sim \mathrm{N}\left(\theta, \sigma^2 \sum_{i=1}^{t} \frac{a_i^2}{r_i}\right).$$

For contrast t-test,

$$t_{\text{obs}} = \frac{\hat{\theta}}{\hat{\sigma}\sqrt{\sum_{i=1}^{t} \frac{a_i^2}{r_i}}} \quad \text{for testing } H_0 : \theta = 0.$$

ANOVA Table

Source	SS	df	Mean Square $= \frac{\text{SS}}{\text{df}}$	F stat
Treatment	$\sum_{i=1}^{t} r_i(\bar{y}_{i.} - \bar{y}_{..})^2$	$t-1$	Trt MS $= \frac{\sum_{i=1}^{t} r_i(\bar{y}_{i.} - \bar{y}_{..})^2}{t-1}$	$F_{\text{obs}} = \frac{\text{Trt MS}}{\hat{\sigma}^2}$
Residuals	$\sum_{i=1}^{t} \sum_{j=1}^{r_i} (y_{ij} - \bar{y}_{i.})^2$	$\sum_{i=1}^{t} r_i - t$	$\hat{\sigma}^2$	
Total	$\sum_{i=1}^{t} \sum_{j=1}^{r_i} (y_{ij} - \bar{y}_{..})^2$	$\sum_{i=1}^{t} r_i - 1$		

To test $H_0 : \tau_1 = \tau_2 = \cdots = \tau_t = 0$, we compare

$$F_{\text{obs}} = \frac{\text{Trt MS}}{\hat{\sigma}^2} \quad \text{to } F_{t-1, \sum_{i=1}^{t} r_i - t}.$$

Example. Imagine some cows got sick, so we have

	Barley	Barley + Lupins	Lupins	Overall
r_i	13	14	11	$n = 38$
Mean	$\bar{y}_{1\cdot} = 3.6$	3.4	3.0	$\bar{y}_{\cdot\cdot} = 3.35$
Var	0.12	0.067	0.1	0.11

We have

- $\hat{\mu} = 3.35$.
- $\hat{\tau}_1 = 3.6 - 3.35 = 0.25$.
- $\hat{\tau}_2 = 3.4 - 3.35 = 0.05$.
- $\hat{\tau}_3 = 3.0 - 3.35 = -0.35$.

Then,

$$\begin{aligned}
\text{Trt SS} &= 13(0.25)^2 + 14(0.05)^2 + 11(-0.35)^2 = 2.195 \\
\text{Tot SS} &= 0.11(38-1) = 4.07 \\
\text{RSS} &= \text{Tot SS} - \text{Trt SS} = 4.07 - 2.195 = 1.875.
\end{aligned}$$

Thus, we have the ANOVA table:

Source	SS	df	Mean Square	F stat
Treatment	2.195	2	1.0975	$F_{\text{obs}} = \frac{1.0975}{0.05357} = 20.5$
Residuals	1.875	35	0.0536	
Total	4.07	37		

Note that F_{obs} is compared to $F_{2,35}$. Using R, we have

```
qf(0.95, 2, 35) = 3.267
```

At level $\alpha = 0.05$, we reject H_0 since $20.5 > 3.267$. There is significant differences among the 3 treatments.

To test $H_0 : \tau_2 - \frac{\tau_1+\tau_3}{2} = 0$, we can use a contrast. The estimate is

$$\hat{\theta} = 0.05 - \frac{0.25 - 0.35}{2} = 0.1.$$

Also, from the ANOVA table, we have

$$\hat{\sigma}^2 = 0.0536.$$

Thus,

$$\text{SE}\left(\hat{\theta}\right) = \sqrt{0.0536}\sqrt{\frac{(-0.5)^2}{13} + \frac{1^2}{14} + \frac{(-0.5)^2}{11}} = 0.078.$$

So,

$$t_{\text{obs}} = \frac{0.1}{0.078} = 1.28 \quad \text{compared to } t_{35}.$$

The rest is left as an exercise.

Lecture 19, 2025/11/12

2.6 Randomized Block Designs (RBD)

The model with t treatments, b blocks, 1 replicate per treatment per block is

$$Y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}, \quad i = 1, 2, \dots, t, \quad j = 1, 2, \dots, b, \quad n = tb.$$

We want μ to represent the overall mean and $\mu + \tau_i$ to represent the mean response in treatment i, which leads to the constraints:

$$\sum_{i=1}^{t} \tau_i = 0 \quad \text{and} \quad \sum_{j=1}^{b} \beta_j = 0$$

where β_j is the effect of block j.

Estimation:

$$\min_{\mu, \tau_i, \beta_j} \sum_{i=1}^{t} \sum_{j=1}^{b} (y_{ij} - \mu - \tau_i - \beta_j)^2 \quad \text{subject to } \sum_{i=1}^{t} \tau_i = 0, \sum_{j=1}^{b} \beta_j = 0$$

gives

$$\hat{\mu} = \bar{y}_{\cdot\cdot}, \quad \hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}, \quad \hat{\beta}_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}.$$

The degree of freedom is

$$\text{df} = tb - (1 + t + b - 2) = tb - t - b + 1 = (t-1)(b-1).$$

The estimate of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{(t-1)(b-1)} \sum_{i=1}^{t} \sum_{j=1}^{b} (\underbrace{y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j}_{y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}})^2.$$

Example. Orange trees in Florida. There are 6 types of fertilizers (treatments) and 7 plots of land (blocks, e.g. different soil quality, amount of sunlight, etc.). We can put 6 trees in each plot. To do RBD,

- 1 tree in each plot gets each fertilizer.
- Randomize positions of treatments in each plot.

Lecture 20, 2025/11/17

Recall the model for RBD:

$$Y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}, \quad i = 1, 2, \ldots, t, \quad j = 1, 2, \ldots, b.$$

- Extension of the 2-treatment model with blocks to multiple treatments, still 1 replicate per block.
- 2 constraints: $\sum_{i=1}^{t} \tau_i = 0$ and $\sum_{j=1}^{b} \beta_j = 0$.
- Assume treatment effect is the same in every block, i.e. we just add $\tau_i + \beta_j$, treatment and block effects simply add up.

Estimation:

$$\hat{\mu} = \bar{y}_{\cdot\cdot}, \quad \hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}, \quad \hat{\beta}_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}.$$

The degree of freedom is $(t-1)(b-1)$ and the estimate of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{(t-1)(b-1)} \sum_{i=1}^{t} \sum_{j=1}^{b} (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2.$$

Key idea: how does blocking change the SS decomposition?

From CRD, we have

$$\underbrace{\sum_{i=1}^{t}\sum_{j=1}^{b}(y_{ij}-\bar{y}_{..})^2}_{\text{Total SS}} = \underbrace{\sum_{i=1}^{t} b(\bar{y}_{i.}-\bar{y}_{..})^2}_{\text{Trt SS}} + \underbrace{\sum_{i=1}^{t}\sum_{j=1}^{b}(y_{ij}-\bar{y}_{i.})^2}_{\text{RSS without blocking}}.$$

With blocking,

$$\underbrace{\sum_{i=1}^{t}\sum_{j=1}^{b}(y_{ij}-\bar{y}_{..})^2}_{\text{Total SS}} = \underbrace{\sum_{i=1}^{t} b(\bar{y}_{i.}-\bar{y}_{..})^2}_{\text{Trt SS}} + ??? + \underbrace{\sum_{i=1}^{t}\sum_{j=1}^{b}(y_{ij}-\bar{y}_{i.}-\bar{y}_{.j}+\bar{y}_{..})^2}_{\text{RSS with blocking}}.$$

We are missing the **Block SS** piece. So blocking separates

$$\text{RSS without blocking} = \text{Block SS} + \text{RSS with blocking}.$$

Now let's decompose the RSS without blocking:

$$\begin{aligned}
\underbrace{\sum_{i=1}^{t}\sum_{j=1}^{b}(y_{ij}-\bar{y}_{i.})^2}_{\text{RSS without blocking}} &= \sum_{i=1}^{t}\sum_{j=1}^{b}\left[(y_{ij}-\bar{y}_{i.}-\bar{y}_{.j}+\bar{y}_{..})+(\bar{y}_{.j}-\bar{y}_{..})\right]^2 \\
&= \underbrace{\sum_{i=1}^{t}\sum_{j=1}^{b}(y_{ij}-\bar{y}_{i.}-\bar{y}_{.j}+\bar{y}_{..})^2}_{\text{RSS with blocking}} + \underbrace{\sum_{j=1}^{b} t(\bar{y}_{.j}-\bar{y}_{..})^2}_{\text{Block SS}} + 2\sum_{i=1}^{t}\sum_{j=1}^{b}(y_{ij}-\bar{y}_{i.}-\bar{y}_{.j}+\bar{y}_{..})(\bar{y}_{.j}-\bar{y}_{..}).
\end{aligned}$$

Note that the last term is

$$2\sum_{j=1}^{b}(\bar{y}_{.j}-\bar{y}_{..})\underbrace{\sum_{i=1}^{t}(y_{ij}-\bar{y}_{i.}-\bar{y}_{.j}+\bar{y}_{..})}_{t\bar{y}_{.j}-t\bar{y}_{..}-t\bar{y}_{.j}+t\bar{y}_{..}=0} = 0.$$

Thus,

$$\underbrace{\sum_{i=1}^{t}\sum_{j=1}^{b}(y_{ij}-\bar{y}_{..})^2}_{\text{Total SS}} = \underbrace{\sum_{i=1}^{t} b(\bar{y}_{i.}-\bar{y}_{..})^2}_{\text{Trt SS}} + \underbrace{\sum_{j=1}^{b} t(\bar{y}_{.j}-\bar{y}_{..})^2}_{\text{Block SS}} + \underbrace{\sum_{i=1}^{t}\sum_{j=1}^{b}(y_{ij}-\bar{y}_{i.}-\bar{y}_{.j}+\bar{y}_{..})^2}_{\text{RSS with blocking}}.$$

We have

$$\frac{\text{RSS}}{\sigma^2} \sim \chi^2_{(t-1)(b-1)}.$$

Also,

$$Y_{ij} \sim \mathrm{N}(\mu + \tau_i + \beta_j, \sigma^2) \implies \bar{Y}_{i\cdot} \sim \mathrm{N}\left(\mu + \tau_i, \frac{\sigma^2}{b}\right).$$

Exercise: Show the distribution of $\bar{Y}_{i\cdot}$.

So for a contrast $\theta = \sum_{i=1}^{t} a_i \tau_i$ with $\sum_{i=1}^{t} a_i = 0$, we have the estimate

$$\hat{\theta} = \sum_{i=1}^{t} a_i \hat{\tau}_i = \sum_{i=1}^{t} a_i \bar{y}_{i\cdot}$$

and

$$\tilde{\theta} \sim \mathrm{N}\left(\theta, \sigma^2 \frac{\sum_{i=1}^{t} a_i^2}{b}\right).$$

So,

$$\frac{\tilde{\theta} - \theta}{\tilde{\sigma}\sqrt{\frac{\sum_{i=1}^{t} a_i^2}{b}}} \sim t_{(t-1)(b-1)}.$$

We use this for contrast t-tests and CIs.

For ANOVA, when $H_0 : \tau_1 = \tau_2 = \cdots = \tau_t = 0$ is true, then,

$$\bar{Y}_{i\cdot} \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{b}\right).$$

So similarly to CRD, we have

$$b\sum_{i=1}^{t}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 \text{ is Trt SS} \quad \text{and} \quad \frac{\text{Trt SS}}{\sigma^2} \sim \chi^2_{(t-1)}$$

and

$$\frac{\text{Trt SS}/(t-1)}{\text{RSS}/((t-1)(b-1))} \sim F_{t-1,(t-1)(b-1)}.$$

We reject H_0 when $F_{\text{obs}} = \frac{\text{Trt MS}}{\hat{\sigma}^2} > c$ where c satisfies $\mathbb{P}(F_{t-1,(t-1)(b-1)} \geq c) = \alpha$ at level α.

ANOVA table for RBD:

Source	SS	df	Mean Square = $\frac{\text{SS}}{\text{df}}$	F stat
Treatment	$b\sum_{i=1}^{t}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$	$t-1$	Trt MS = $\frac{b\sum_{i=1}^{t}(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})^2}{t-1}$	$F_{\text{obs}} = \frac{\text{Trt MS}}{\hat{\sigma}^2}$
Blocks	$t\sum_{j=1}^{b}(\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2$	$b-1$	Block MS = $\frac{t\sum_{j=1}^{b}(\bar{y}_{\cdot j}-\bar{y}_{\cdot\cdot})^2}{b-1}$	
Residuals	$\sum_{i=1}^{t}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2$	$(t-1)(b-1)$	$\hat{\sigma}^2$	
Total	$\sum_{i=1}^{t}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{\cdot\cdot})^2$	$tb-1$		

Lecture 21, 2025/11/19

2.7 Factorial Design with 2 Factors

Consider the t treatment model:

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}, \quad i = 1, 2, \dots, t, \quad j = 1, 2, \dots, r$$

where treatments are combinations of factors A and B. That is,

$$t = t_a \times t_b$$

where t_a = # levels of A and t_b = # levels of B.

The t-treatment model has parameters τ_i for <u>every</u> combination of factors A and B.

$$\text{\# of free parameters} = \underbrace{1}_{\mu} + \underbrace{(t-1)}_{\tau_i \text{ with } \sum \tau_i = 0}.$$

If the levels of factor A have the same effect on each level of factor B, then we say their effects are **additive**.

Define:

- α_i = effect of level i of Factor A.
- β_j = effect of level j of Factor B.

If effects are not additive, then we say Factor A and B have **interaction**.

The additive model requires fewer parameters:

Example. $t_a = 2, t_b = 3$, so $t = 6$ treatments.

Factor			Mean	
A	B	Treatment	General t-Treatment Model	Additive Model
1	1	1	$\mu + \tau_1$	$\mu + \alpha_1 + \beta_1$
1	2	2	$\mu + \tau_2$	$\mu + \alpha_1 + \beta_2$
1	3	3	$\mu + \tau_3$	$\mu + \alpha_1 + \beta_3$
2	1	4	$\mu + \tau_4$	$\mu + \alpha_2 + \beta_1$
2	2	5	$\mu + \tau_5$	$\mu + \alpha_2 + \beta_2$
2	3	6	$\mu + \tau_6$	$\mu + \alpha_2 + \beta_3$

	General t-Treatment Model	Additive Model
Constraints	$\sum_{i=1}^{6} \tau_i = 0$	$\begin{cases} \alpha_1 + \alpha_2 = 0 \\ \beta_1 + \beta_2 + \beta_3 = 0 \end{cases}$
# of Free Parameters	$1 + (6 - 1) = 6$	$1 + (2 - 1) + (3 - 1) = 4$

In this example, no interaction between factors A and B means that

$$\tau_4 - \tau_1 = \tau_5 - \tau_2 = \tau_6 - \tau_3 = \alpha_2 - \alpha_1.$$

Example. Consider fries with two factors:

Factor A	Factor B
1 = no ketchup	1 = no vinegar
2 = ketchup	2 = vinegar

If the additive model is appropriate, suppose response = "average tastiness":

- With ketchup > no ketchup: $8 > 6$.
- With vinegar > no vinegar: $7 > 6$.
- With both ketchup and vinegar is as tasty as $(8 - 6) + (7 - 6) + 6 = 9$. If that is not possible, then 2 factors interact.

Question: Is there interaction between factors A and B?

- If not, then additive model would be appropriate.
- If yes, then general t-treatment model is more appropriate.

Example. Visualizing interaction between two factors example in slides.

When plotting the response vs. one factor, if the lines are parallel, then there is no interaction.

2.8 Two Factors with Interaction

Rewrite CRD with t treatments and r replicates per treatment (i.e. $Y_{ij} = \mu + \tau_i + \epsilon_{ij}$) with interaction and additive effects:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \epsilon_{ijk}, \quad \begin{cases} i = 1, 2, \dots, t_a \\ j = 1, 2, \dots, t_b \\ k = 1, 2, \dots, r \end{cases}$$

where

- μ is the overall mean.
- α_i is the main/additive effect of level i of Factor A.
- β_j is the main/additive effect of level j of Factor B.
- γ_{ij} is the interaction effect between level i of Factor A and level j of Factor B.
- r is the # of replicates per treatment.
- $\epsilon_{ijk} \sim \mathrm{N}(0, \sigma^2)$ are all independent.
- Total sample size is $tr = t_a t_b r$, where $t = t_a t_b$ is the total # of treatments.

Constraints: For μ to represent the overall mean, and α_i's and β_j's to represent additive effects, etc. We need the following constraints:

$$\sum_{i=1}^{t_a} \alpha_i = 0, \quad \sum_{j=1}^{t_b} \beta_j = 0, \quad \sum_{i=1}^{t_a} \gamma_{ij} = 0 \ \forall j, \quad \sum_{j=1}^{t_b} \gamma_{ij} = 0 \ \forall i.$$

Free parameters:

- $t - 1$ free parameters among τ_i's in the equivalent CRD t-treatment model. So, it must equal the # of free parameters in α_i's, β_j's, and γ_{ij}'s.
- $t_a - 1$ free parameters in α_i's.

- $t_b - 1$ free parameters in β_j's.
- So $t - 1 - (t_a - 1) - (t_b - 1) = t_a t_b - t_a - t_b + 1 = (t_a - 1)(t_b - 1)$ free parameters in γ_{ij}'s.

Estimation:

$$\min_{\mu,\alpha_i,\beta_j,\gamma_{ij}} \sum_{i=1}^{t_a}\sum_{j=1}^{t_b}\sum_{k=1}^{r}(y_{ijk} - \mu - \alpha_i - \beta_j - \gamma_{ij})^2 \quad \text{subject to constraints above.}$$

This gives

$$\hat{\mu} = \bar{y}_{...}, \quad \hat{\alpha}_i = \bar{y}_{i..} - \bar{y}_{...}, \quad \hat{\beta}_j = \bar{y}_{.j.} - \bar{y}_{...}, \quad \hat{\gamma}_{ij} = \bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}.$$

Then, the estimate of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{t_a t_b (r-1)} \sum_{i=1}^{t_a}\sum_{j=1}^{t_b}\sum_{k=1}^{r}(\underbrace{y_{ijk} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\gamma}_{ij}}_{y_{ijk} - \bar{y}_{ij.}})^2 \quad \text{same as CRD } t\text{-treatment.}$$

ANOVA Decomposition:

We know from CRD that

$$\text{RSS} = \sum_{i,j,k}(y_{ijk} - \bar{y}_{ij.})^2 \quad \text{and} \quad \text{Trt SS} = r\sum_{i,j}(\bar{y}_{ij.} - \bar{y}_{...})^2.$$

Decompose Trt SS in terms of A, B, and interaction:

$$\underbrace{r\sum_{i,j}(\bar{y}_{ij.} - \bar{y}_{...})^2}_{\text{Trt SS}} = \underbrace{r\sum_{i,j}(\bar{y}_{i..} - \bar{y}_{...})^2}_{\text{SS}_A} + \underbrace{r\sum_{i,j}(\bar{y}_{.j.} - \bar{y}_{...})^2}_{\text{SS}_B} + \underbrace{r\sum_{i,j}(\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2}_{\text{SS}_I}.$$

This holds from RBD decomposition by ignoring "r" and the third $\cdot$ index.

ANOVA Table:

1) Separate t-treatment CRD.

Source	SS	df	Mean Square $= \frac{\text{SS}}{\text{df}}$	F stat
Treatment	$r\sum_{i,j}(\bar{y}_{ij.} - \bar{y}_{...})^2$	$t_a t_b - 1$	Trt MS $= \frac{r\sum_{i,j}(\bar{y}_{ij.} - \bar{y}_{...})^2}{t_a t_b - 1}$	$F_{\text{obs}} = \frac{\text{Trt MS}}{\hat{\sigma}^2}$
Residuals	$\sum_{i,j,k}(y_{ijk} - \bar{y}_{ij.})^2$	$t_a t_b (r-1)$	$\hat{\sigma}^2$	
Total	$\sum_{i,j,k}(y_{ijk} - \bar{y}_{...})^2$	$t_a t_b r - 1$		

Use the F_{obs} to test for differences among the t treatments, compared to $F_{t_a t_b - 1, t_a t_b (r-1)}$.

2) With interactions.

Source	SS	df	Mean Square $= \frac{\text{SS}}{\text{df}}$	F stat
Factor A	SS_A	$t_a - 1$	MS_A	$F_{A,\text{obs}} = \frac{\text{MS}_A}{\hat{\sigma}^2}$
Factor B	SS_B	$t_b - 1$	MS_B	$F_{B,\text{obs}} = \frac{\text{MS}_B}{\hat{\sigma}^2}$
Interaction	SS_I	$(t_a - 1)(t_b - 1)$	MS_I	$F_{I,\text{obs}} = \frac{\text{MS}_I}{\hat{\sigma}^2}$
Residuals	RSS	$t_a t_b (r-1)$	$\hat{\sigma}^2$	
Total	Total SS	$t_a t_b r - 1$		

Use $F_{A,\text{obs}}$, $F_{B,\text{obs}}$, and $F_{I,\text{obs}}$ to test for significance of Factor A, Factor B, and interaction, respectively. Compare to $F_{t_a - 1, t_a t_b (r-1)}$, $F_{t_b - 1, t_a t_b (r-1)}$, and $F_{(t_a - 1)(t_b - 1), t_a t_b (r-1)}$ respectively.

Lecture 23, 2025/11/26

Example. Factor design with 2 factors example in slides.

2.9 Two Factors with Blocks

Consider t treatments and b blocks, 1 replicate per treatment per block. We will re-express RBD model in terms of additive and interaction effects, where $t = t_a t_b$ are treatments as a combination of t_a levels of Factor A and t_b levels of Factor B. The model is

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \lambda_k + \epsilon_{ijk}, \quad \begin{cases} i = 1, 2, \ldots, t_a \\ j = 1, 2, \ldots, t_b \\ k = 1, 2, \ldots, b \end{cases}$$

where the parameters have the same interpretations as in the two factor interaction model, with λ_k being the block effects.

As before, block effects are assumed to be additive and they don't interact with either factor.

Constraints:

$$\sum_{i=1}^{t_a} \alpha_i = 0, \quad \sum_{j=1}^{t_b} \beta_j = 0, \quad \sum_{i=1}^{t_a} \gamma_{ij} = 0 \ \forall j, \quad \sum_{j=1}^{t_b} \gamma_{ij} = 0 \ \forall i, \quad \sum_{k=1}^{b} \lambda_k = 0 \text{ (from RBD)}.$$

Estimation:

$$\min_{\mu,\alpha_i,\beta_j,\gamma_{ij},\lambda_k} \sum_{i=1}^{t_a}\sum_{j=1}^{t_b}\sum_{k=1}^{b}(y_{ijk}-\mu-\alpha_i-\beta_j-\gamma_{ij}-\lambda_k)^2 \quad \text{subject to constraints above.}$$

This gives

$$\hat{\mu}=\bar{y}_{...}, \quad \hat{\alpha}_i=\bar{y}_{i..}-\bar{y}_{...}, \quad \hat{\beta}_j=\bar{y}_{.j.}-\bar{y}_{...}, \quad \hat{\gamma}_{ij}=\bar{y}_{ij.}-\bar{y}_{i..}-\bar{y}_{.j.}+\bar{y}_{...}, \quad \hat{\lambda}_k=\bar{y}_{..k}-\bar{y}_{...}.$$

The residual degree of freedom is

$$\mathrm{df}=(t-1)(b-1)=(t_at_b-1)(b-1) \quad \text{from RBD.}$$

The estimate of σ^2 is

$$\hat{\sigma}^2=\frac{1}{(t_at_b-1)(b-1)}\sum_{i=1}^{t_a}\sum_{j=1}^{t_b}\sum_{k=1}^{b}(\underbrace{y_{ijk}-\hat{\mu}-\hat{\alpha}_i-\hat{\beta}_j-\hat{\gamma}_{ij}-\hat{\lambda}_k}_{y_{ijk}-\bar{y}_{ij.}-\bar{y}_{..k}+\bar{y}_{...}})^2 \quad \text{same as RBD.}$$

SS Decomposition: We have

$$\text{Trt SS}=b\sum_{i,j}(\bar{y}_{ij.}-\bar{y}_{...})^2=\underbrace{b\sum_{i,j}(\bar{y}_{i..}-\bar{y}_{...})^2}_{\mathrm{SS}_A}+\underbrace{bt_a\sum_{i,j}(\bar{y}_{.j.}-\bar{y}_{...})^2}_{\mathrm{SS}_B}+\underbrace{b\sum_{i,j}(\bar{y}_{ij.}-\bar{y}_{i..}-\bar{y}_{.j.}+\bar{y}_{...})^2}_{\mathrm{SS}_I}.$$

Also, from RBD, we have

$$\text{Block SS}=t_at_b\sum_{k=1}^{b}(\bar{y}_{..k}-\bar{y}_{...})^2, \quad \text{RSS}=\sum_{i=1}^{t_a}\sum_{j=1}^{t_b}\sum_{k=1}^{b}(y_{ijk}-\bar{y}_{ij.}-\bar{y}_{..k}+\bar{y}_{...})^2.$$

ANOVA Table:

1) Separate treatments.

Source	SS	df	Mean Square $=\frac{\mathrm{SS}}{\mathrm{df}}$	F stat
Treatment	$b\sum_{i,j}(\bar{y}_{ij.}-\bar{y}_{...})^2$	t_at_b-1	Trt MS $=\frac{b\sum_{i,j}(\bar{y}_{ij.}-\bar{y}_{...})^2}{t_at_b-1}$	$F_{\text{obs}}=\frac{\text{Trt MS}}{\hat{\sigma}^2}$
Blocks	$t_at_b\sum_{k=1}^{b}(\bar{y}_{..k}-\bar{y}_{...})^2$	$b-1$	Block MS $=\frac{t_at_b\sum_{k=1}^{b}(\bar{y}_{..k}-\bar{y}_{...})^2}{b-1}$	
Residuals	$\sum_{i,j,k}(y_{ijk}-\bar{y}_{ij.}-\bar{y}_{..k}+\bar{y}_{...})^2$	$(t_at_b-1)(b-1)$	$\hat{\sigma}^2$	
Total	$\sum_{i,j,k}(y_{ijk}-\bar{y}_{...})^2$	t_at_bb-1		

2) With Factor A, B, and interaction.

Source	SS	df	Mean Square $= \frac{\text{SS}}{\text{df}}$	F stat
Factor A	SS_A	$t_a - 1$	MS_A	$F_{A,\text{obs}} = \frac{\text{MS}_A}{\hat{\sigma}^2}$
Factor B	SS_B	$t_b - 1$	MS_B	$F_{B,\text{obs}} = \frac{\text{MS}_B}{\hat{\sigma}^2}$
Interaction	SS_I	$(t_a - 1)(t_b - 1)$	MS_I	$F_{I,\text{obs}} = \frac{\text{MS}_I}{\hat{\sigma}^2}$
Blocks	Block SS	$b - 1$	Block MS	
Residuals	RSS	$(t_a t_b - 1)(b - 1)$	$\hat{\sigma}^2$	
Total	Total SS	$t_a t_b b - 1$		

Lecture 24, 2025/12/01

Possible hypothesis tests:

- Trt SS $= \text{SS}_A + \text{SS}_B + \text{SS}_I$. Compare $\frac{\text{Trt MS}}{\hat{\sigma}^2}$ to $F_{t_a t_b - 1, (t_a t_b - 1)(b-1)}$ to test for any differences among the treatments.
- Compare $\frac{\text{MS}_A}{\hat{\sigma}^2}$ to $F_{t_a - 1, (t_a t_b - 1)(b-1)}$ to test for significance of Factor A.
- Compare $\frac{\text{MS}_B}{\hat{\sigma}^2}$ to $F_{t_b - 1, (t_a t_b - 1)(b-1)}$ to test for significance of Factor B.
- Compare $\frac{\text{MS}_I}{\hat{\sigma}^2}$ to $F_{(t_a - 1)(t_b - 1), (t_a t_b - 1)(b-1)}$ to test if there are any significant interactions between Factors A and B.

Interaction Workflow

- Are any of the treatments significantly different? (Use $F = \frac{\text{Trt MS}}{\hat{\sigma}^2}$)
  - If no, done.
  - If yes, are there significant interactions? (Use $F = \frac{\text{MS}_I}{\hat{\sigma}^2}$)
    * If no, use model with additive effects. Then, SS_I gets added to RSS, set $\gamma_{ij} = 0$, and check if Factors A and B are individually significant, know their effects just add together.
    * If yes, make interaction plots to visualize.

Example. Two factors with blocks example in slides.

Be familiar with the following R commands for ANOVA.

```
## All the variants of the "aov" command:
summary(aov(y ~ trt))
summary(aov(y ~ trt + block))
summary(aov(y ~ A:B))
summary(aov(y ~ A*B))
summary(aov(y ~ A+B))
summary(aov(y ~ A:B + block))
summary(aov(y ~ A*B + block))
```

END OF STAT 332!